

CEH- Module 1: Introduction to Ethical Hacking

- Information Security Overview
- Hacking Concepts
- Ethical Hacking Concepts
- Hacking Methodologies and Frameworks
- Information Security Controls
- Information Security Laws and Standards

Module 2: Footprinting and Reconnaissance

- Footprinting Concepts
- Footprinting through Search Engines
- Footprinting through Internet Research Services
- Footprinting through Social Networking Sites
- Whois Footprinting
- DNS Footprinting
- Network and Email Footprinting
- Footprinting through Social Engineering
- Footprinting Tasks using Advanced Tools and AI
- Footprinting Countermeasures

Module 3: Scanning Networks

- Network Scanning Concepts
- Scanning Tools
- Host Discovery
- Port and Service Discovery
- OS Discovery (Banner Grabbing/OS Fingerprinting)
- Scanning Beyond IDS and Firewall
- Source Port Manipulation
- Network Scanning Countermeasures

Module 4: Enumeration

- Enumeration Concepts
- NetBIOS Enumeration
- SNMP Enumeration
- LDAP Enumeration
- NTP and NFS Enumeration
- SMTP and DNS Enumeration
- Other Enumeration Techniques
- Enumeration Countermeasures

Module 5: Vulnerability Analysis

- Vulnerability Assessment Concepts
- Vulnerability Scoring Systems and Databases
- Vulnerability-Management Life Cycle
- Vulnerability Research
- Vulnerability Scanning and Analysis
- Vulnerability Assessment Tools
- Vulnerability Assessment Reports

Module 6: System Hacking

- Gaining Access
- Escalating Privileges
- Maintaining Access
- Clearing Logs

Module 7: Malware Threats

- Malware Concepts
- APT Concepts

- Trojan Concepts
- Viruses and Worms
- Fileless Malware Concepts
- AI-based Malware Concepts
- Malware Countermeasures
- Anti-Malware Software

Module 8: Sniffing

- Sniffing Concepts
- Sniffing Technique
- Sniffing Tools
- Sniffing Countermeasures

Module 9: Social Engineering

- Social Engineering Concepts
- Human-based Social Engineering Techniques
- Computer-based Social Engineering Techniques
- Mobile-based Social Engineering Techniques
- Social Engineering Countermeasures

Module 10: Denial-of-Service

- DoS/DDoS Concepts
- Botnets
- DDoS Case Study
- DoS/DDoS Attack Techniques
- DoS/DDoS Attack Countermeasures

Module 11: Session Hijacking

- Session Hijacking Concepts

- Application-Level Session Hijacking
- Network-Level Session Hijacking
- Session Hijacking Tools
- Session Hijacking Countermeasures

Module 12: Evading IDS, Firewalls, and Honeypots

- Intrusion Detection/Prevention System (IDS/IPS) Concepts
- Firewall Concepts
- IDS, IPS, and Firewall Solutions
- Evading IDS/Firewalls
- Evading NAC and Endpoint Security
- IDS/Firewall Evading Tools
- Honeypot Concepts
- IDS/Firewall Evasion Countermeasures

Module 13: Hacking Web Servers

- Web Server Concepts
- Web Server Attacks
- Web Server Attack Methodology
- Web Server Attack Countermeasures
- Patch Management

Module 14: Hacking Web Applications

- Web Application Concepts
- Web Application Threats
- Web Application Hacking Methodology
- Web API and Webhooks
- Web Application Security

Module 15: SQL Injections

- SQL Injection Concepts
- Types of SQL Injection
- SQL Injection Methodology
- Evasion Techniques
- SQL Injection Countermeasures

Module 16: Hacking Wireless Networks

- Wireless Concepts
- Wireless Encryption
- Wireless Threats
- Wireless Hacking Methodology
- Wireless Attack Countermeasures

Module 17: Hacking Mobile Platforms

- Mobile Platform Attack Vectors
- Hacking Android OS
- Hacking iOS
- Mobile Device Management
- Mobile Security Guidelines
- Mobile Security Tools

Module 18: IoT Hacking & OT Hacking

- IoT Hacking
- IoT Concepts and Attacks
- IoT Hacking Methodology
- OT Hacking

Module 19: Cloud Computing

- Cloud Computing Concepts
- Container Technology
- Serverless Computing
- Cloud Hacking
- Cloud Hacking Methodology
- AWS Hacking
- Microsoft Azure Hacking
- Google Cloud Hacking
- Container Hacking
- Cloud Security

Module 20: Cryptography

- Cryptography Concepts and Encryption Algorithms
- Applications of Cryptography
- Cryptanalysis
- Cryptography Attack Countermeasures

CND- The courseware of CNDv2 is designed to develop a strong and deep understanding of various networking verticals, talking first about various network and defense strategies proceeding to incident detection and response and advancing to threat assessments and intelligence. Below is the list of modules with a brief description of what they talk about.

Module 01: Network Attacks and Defense Strategies: This module introduces you to different network-based attacks faced by the organization to understand their working and develop defense strategies.

Module 02: Administrative Network Security: It involves developing or updating security infrastructure and continuously monitoring networks for any suspicious actions or unauthorized access

Module 03: Technical Network Security: Implementing authentication and protection controls for user verification to avoid theft of sensitive information or data.

Introducing the concept of zero trust and its effectiveness in maintaining a better security posture

Module 04: Network Perimeter Security: Implementation and management of perimeter devices like firewalls, Intrusion Detection Systems, Intrusion Prevention Systems

Module 05: Endpoint Security-Windows Systems: Security of end-user devices and entry points by implying endpoint security on Windows devices.

Module 06: Endpoint Security-Linux Systems: Securing entry points or end-user devices by ensuring endpoint security on Linux devices

Module 07: Endpoint Security- Mobile Devices: Securing entry points or end-user devices by ensuring endpoint security on mobile devices

Module 08: Endpoint Security-IoT Devices: Fundamentals of IoT, IoT threats and security using endpoint security implementation

Module 09: Administrative Application Security: Understanding the methodologies of administrative application security and its importance to minimize the security-related vulnerabilities in the application

Module 10: Data Security: Implementing policies to safeguard data from unauthorized access using various techniques like encryption, hashing, tokenization, and other key management practices. Concept of data storage, data classification, data masking, retention and destruction

Module 11: Enterprise Virtual Network Security: In-depth understanding of virtualization, related threats, and security. Essentials of software-defined network (SDN) security, network function virtualization (NFV) security

Module 12: Enterprise Cloud Network Security: Introduction to cloud computing, threats, challenges and security across cloud platforms, concepts of container security, docker security, and Kubernetes security

Module 13: Enterprise Wireless Network Security: Understanding of wireless network security essentials, threats, attacks, and countermeasures.

Module 14: Network Traffic Monitoring and Analysis: Analysis and monitoring of logs from various perimeter network devices to identify any anomalies in the traffic.

Module 15: Network Logs Monitoring and Analysis: Analyzing the events generated by various devices in the network to identify signs of any suspicious activity or a potential incident

Module 16: Incident Response and Forensic Investigations: Understanding of incident management response process and methodologies to be followed in case of security incidents. Understanding of forensics investigation techniques and tools used for analysis.

Module 17: Business Continuity and Disaster Recovery: Understanding the importance of BCP and DR, related concepts and procedures required to allow smooth functioning of operations in case of a disaster

Module 18: Risk Anticipation with Risk Management: Risk management process, analyzing various risks that the organization is susceptible to and developing policies to manage them.

Module 19: Threat Assessment with Attack Surface Analysis: Analyzing the threats and attack vectors to develop solutions for their countermeasures

Module 20: Threat Prediction with Cyber Threat Intelligence: Developing a proactive approach by understanding various frameworks aiding in threat intelligence to anticipate the kinds of attacks hackers could use to gain access to the network.

- CIH- **Module 01:** Introduction to Incident Handling and Response
- **Module 02:** Incident Handling and Response Process
- **Module 03:** Forensic Readiness and First Response
- **Module 04:** Handling and Responding to Malware Incidents
- **Module 05:** Handling and Responding to Email Security Incidents
- **Module 06:** Handling and Responding to Network Security Incidents
- **Module 07:** Handling and Responding to Web Application Security Incidents
- **Module 08:** Handling and Responding to Cloud Security Incidents
- **Module 09:** Handling and Responding to Insider Threats

CPENT- CPENT Certification Training Course consists of 14 modules and tests the abilities of a penetration tester in almost all the vectors of cybersecurity, some of which have been introduced for the first time in any penetration certification

- **Module 01:** Introduction to Penetration Testing
- **Module 02:** Penetration Testing Scoping and Engagement
- **Module 03:** Open Source Intelligence (OSINT)
- **Module 04:** Social Engineering Penetration Testing
- **Module 05:** Network Penetration Testing – External
- **Module 06:** Network Penetration Testing– Internal

- **Module 07:** Network Penetration Testing – Perimeter Devices
- **Module 08:** Web Application Penetration Testing
- **Module 09:** Wireless Penetration Testing
- **Module 10:** IoT Penetration Testing
- **Module 11:** OT/SCADA Penetration Testing
- **Module 12:** Cloud Penetration Testing
- **Module 13:** Binary Analysis and Exploitation
- **Module 14:** Report Writing and Post Testing Actions

CISSP - Domain 1: Security and Risk Management

- **1.1 – Understand, adhere to, and promote professional ethics**
 - ISC2 Code of Professional Ethics
 - Organizational code of ethics
- **1.2 – Understand and apply security concepts**
 - Confidentiality, integrity, and availability, authenticity, and nonrepudiation (5 Pillars of Information Security)
- **1.3 – Evaluate and apply security governance principles**
 - Alignment of the security function to business strategy, goals, mission, and objectives
 - Organizational processes (e.g., acquisitions, divestitures, governance committees)
 - Organizational roles and responsibilities
 - Security control frameworks (e.g., International Organization for Standardization (ISO), National Institute of Standards and Technology (NIST), Control Objectives for Information and Related Technology (COBIT), Sherwood Applied Business Security Architecture (SABSA), Payment Card Industry (PCI), Federal Risk and Authorization Management Program (FedRAMP))

- Due care/due diligence
- **1.4 – Understand legal, regulatory, and compliance issues that pertain to information security in a holistic context**
 - Cybercrimes and data breaches
 - Licensing and Intellectual Property requirements
 - Import/export controls
 - Transborder data flow
 - Issues related to privacy (e.g., General Data Protection Regulation (GDPR), California Consumer Privacy Act, Personal Information Protection Law, Protection of Personal Information Act)
 - Contractual, legal, industry standards, and regulatory requirements
- **1.5 – Understand requirements for investigation types (i.e., administrative, criminal, civil, regulatory, industry standards)**
- **1.6 – Develop, document, and implement security policy, standards, procedures, and guidelines**
- **1.7 – Identify, analyze, assess, prioritize, and implement Business Continuity (BC) requirements**
 - Business impact analysis (BIA)
 - External dependencies
- **1.8 – Contribute to and enforce personnel security policies and procedures**
 - Candidate screening and hiring
 - Employment agreements and policy driven requirements
 - Onboarding, transfers, and termination processes
 - Vendor, consultant, and contractor agreements and controls
- **1.9 – Understand and apply risk management concepts**
 - Threat and vulnerability identification
 - Risk analysis, assessment, and scope
 - Risk response and treatment (e.g., cybersecurity insurance)
 - Applicable types of controls (e.g., preventive, detection, corrective)
 - Control assessments (e.g., security and privacy)

- Continuous monitoring and measurement
- Reporting (e.g., internal, external)
- Continuous improvement (e.g., risk maturity modeling)
- Risk frameworks (e.g., International Organization for Standardization (ISO), National Institute of Standards and Technology (NIST), Control Objectives for Information and Related Technology (COBIT), Sherwood Applied Business Security Architecture (SABSA), Payment Card Industry (PCI))
- **1.10 – Understand and apply threat modeling concepts and methodologies**
- **1.11 – Apply Supply Chain Risk Management (SCRM) concepts**
 - Risks associated with the acquisition of products and services from suppliers and providers (e.g., product tampering, counterfeits, implants)
 - Risk mitigations (e.g., third-party assessment and monitoring, minimum security requirements, service level requirements, silicon root of trust, physically unclonable function, software bill of materials)
- **1.12 – Establish and maintain a security awareness, education, and training program**
 - Methods and techniques to increase awareness and training (e.g., social engineering, phishing, security champions, gamification)
 - Periodic content reviews to include emerging technologies and trends (e.g., cryptocurrency, artificial intelligence (AI), blockchain)
 - Program effectiveness evaluation

Domain 2: Asset Security

- **2.1 – Identify and classify information and assets**
 - Data classification
 - Asset Classification
- **2.2 – Establish information and asset handling requirements**
- **2.3 – Provision information and assets securely**
 - Information and asset ownership
 - Asset inventory (e.g., tangible, intangible)
 - Asset management
- **2.4 – Manage data lifecycle**

- Data roles (i.e., owners, controllers, custodians, processors, users/subjects)
- Data collection
- Data location
- Data maintenance
- Data retention
- Data remanence
- Data destruction
- **2.5 – Ensure appropriate asset retention (e.g., End of Life (EOL), End of Support)**
- **2.6 – Determine data security controls and compliance requirements**
 - Data states (e.g., in use, in transit, at rest)
 - Scoping and tailoring
 - Standards selection
 - Data protection methods (e.g., Digital Rights Management (DRM), Data Loss Prevention (DLP), Cloud Access Security Broker (CASB))

Domain 3: Security Architecture and Engineering

- **3.1 – Research, implement and manage engineering processes using secure design principles**
 - Threat modeling
 - Least privilege
 - Defense in depth
 - Secure defaults
 - Fail securely
 - Segregation of Duties (SoD)
 - Keep it simple and small
 - Zero trust or trust but verify
 - Privacy by design
 - Shared responsibility
 - Secure access service edge

- **3.2 – Understand the fundamental concepts of security models (e.g., Biba, Star Model, Bell-LaPadula)**
- **3.3 – Select controls based upon systems security requirements**
- **3.4 – Understand security capabilities of Information Systems (IS) (e.g., memory protection, Trusted Platform Module (TPM), encryption/decryption)**
- **3.5 – Assess and mitigate the vulnerabilities of security architectures, designs, and solution elements**
 - Client-based systems
 - Server-based systems
 - Database systems
 - Cryptographic systems
 - Industrial Control Systems (ICS)
 - Cloud-based systems (e.g., Software as a Service (SaaS), Infrastructure as a Service (IaaS), Platform as a Service (PaaS))
 - Distributed systems
 - Internet of Things (IoT)
 - Microservices (e.g., application programming interface (API))
 - Containerization
 - Serverless
 - Embedded systems
 - High-Performance Computing systems
 - Edge computing systems
 - Virtualized systems
- **3.6 – Select and determine cryptographic solutions**
 - Cryptographic life cycle (e.g., keys, algorithm selection)
 - Cryptographic methods (e.g., symmetric, asymmetric, elliptic curves, quantum)
 - Public key infrastructure (PKI) (e.g., quantum key distribution)
- **3.7 – Understand methods of cryptanalytic attacks**
 - Brute force

- Ciphertext only
- Known plaintext
- Frequency analysis
- Chosen ciphertext
- Implementation attacks
- Side-channel
- Fault injection
- Timing
- Man-in-the-Middle (MITM)
- Pass the hash
- Kerberos exploitation
- Ransomware
- **3.8 – Apply security principles to site and facility design**
- **3.9 – Design site and facility security controls**
 - Wiring closets/intermediate distribution facilities
 - Server rooms/data centers
 - Media storage facilities
 - Evidence storage
 - Restricted and work area security
 - Utilities and Heating, Ventilation, and Air Conditioning (HVAC)
 - Environmental issues (e.g., natural disasters, man-made)
 - Fire prevention, detection, and suppression
 - Power (e.g., redundant, backup)
- **3.10 – Manage the information system lifecycle**
 - Stakeholders needs and requirements
 - Requirements analysis
 - Architectural design
 - Development /implementation
 - Integration

- Verification and validation
- Transition/deployment
- Operations and maintenance/sustainment
- Retirement/disposal

Domain 4: Communication and Network Security

• 4.1 – Apply secure design principles in network architectures

- Open System Interconnection (OSI) and Transmission Control Protocol/Internet Protocol (TCP/IP) models
- Internet Protocol (IP) version 4 and 6 (IPv6) (e.g., unicast, broadcast, multicast, anycast)
- Secure protocols (e.g., Internet Protocol Security (IPSec), Secure Shell (SSH), Secure Sockets Layer (SSL)/ Transport Layer Security (TLS))
- Implications of multilayer protocols
- Converged protocols (e.g., Internet Small Computer Systems Interface (iSCSI), Voice over Internet Protocol (VoIP), InfiniBand over Ethernet, Compute Express Link)
- Transport architecture (e.g., topology, data/control/management plane, cut-through/store-and-forward)
- Performance metrics (e.g., bandwidth, latency, jitter, throughput, signal-to-noise ratio)
- Traffic flows (e.g., north-south, east-west)
- Physical segmentation (e.g., in-band, out-of-band, air-gapped)
- Logical segmentation (e.g., virtual local area networks (VLANs), virtual private networks (VPNs), virtual routing and forwarding, virtual domain)
- Micro-segmentation (e.g., network overlays/encapsulation; distributed firewalls, routers, intrusion detection system (IDS)/intrusion prevention system (IPS), zero trust)
- Edge networks (e.g., ingress/egress, peering)
- Wireless networks (e.g., Bluetooth, Wi-Fi, Zigbee, satellite)
- Cellular/mobile networks (e.g., 4G, 5G)
- Content distribution networks (CDN)

- Software defined networks (SDN), (e.g., application programming interface (API), Software-Defined Wide- Area Network, network functions virtualization)
- Virtual Private Cloud (VPC)
- Monitoring and management (e.g., network observability, traffic flow/shaping, capacity management, fault detection and handling)
- **4.2 – Secure network components**
 - Operation of infrastructure (e.g., redundant power, warranty, support)
 - Transmission media (e.g., physical security of media, signal propagation quality)
 - Network Access Control (NAC) systems (e.g., physical, and virtual solutions)
 - Endpoint security (e.g., host-based)
- **4.3 – Implement secure communication channels according to design**
 - Voice, video, and collaboration (e.g., conferencing, Zoom rooms)
 - Remote access (e.g., network administrative functions)
 - Data communications (e.g., backhaul networks, satellite)
 - Third-party connectivity (e.g., telecom providers, hardware support)

Domain 5: Identity and Access Management (IAM)

- **5.1 – Control physical and logical access to assets**
 - Information
 - Systems
 - Devices
 - Facilities
 - Applications
 - Services
- **5.2 – Design identification and authentication strategy (e.g., people, devices, and services)**
 - Groups and Roles
 - Authentication, Authorization and Accounting (AAA) (e.g., multi-factor authentication (MFA), password-less authentication)

- Session management
- Registration, proofing, and establishment of identity
- Federated Identity Management (FIM)
- Credential management systems (e.g., Password vault)
- Single sign-on (SSO)
- Just-In-Time
- **5.3 – Federated identity with a third-party service**
 - On-premise
 - Cloud
 - Hybrid
- **5.4 – Implement and manage authorization mechanisms**
 - Role-based access control (RBAC)
 - Rule based access control
 - Mandatory access control (MAC)
 - Discretionary access control (DAC)
 - Attribute-based access control (ABAC)
 - Risk based access control
 - Access policy enforcement (e.g., policy decision point, policy enforcement point)
- **5.5 – Manage the identity and access provisioning lifecycle**
 - Account access review (e.g., user, system, service)
 - Provisioning and deprovisioning (e.g., on /off boarding and transfers)
 - Role definition and transition (e.g., people assigned to new roles)
 - Privilege escalation (e.g., use of sudo, auditing its use)
 - Service accounts management
- **5.6 – Implement authentication systems**

Domain 6: Security Assessment and Testing

- **6.1 – Design and validate assessment, test, and audit strategies**
 - Internal (e.g., within organization control)

- External (e.g., outside organization control)
- Third-party (e.g., outside of enterprise control)
- Location (e.g., on-premises, cloud, hybrid)
- **6.2 – Conduct security control testing**
 - Vulnerability assessment
 - Penetration testing (e.g., red, blue, and/or purple team exercises)
 - Log reviews
 - Synthetic transactions/benchmarks
 - Code review and testing
 - Misuse case testing
 - Coverage analysis
 - Interface testing (e.g., user interface, network interface, application programming interface (API))
 - Breach attack simulations
 - Compliance checks
- **6.3 – Collect security process data (e.g., technical and administrative)**
 - Account management
 - Management review and approval
 - Key performance and risk indicators
 - Backup verification data
 - Training and awareness
 - Disaster Recovery (DR) and Business Continuity (BC)
- **6.4 – Analyze test output and generate report**
 - Remediation
 - Exception handling
 - Ethical disclosure
- **6.5 – Conduct or facilitate security audits**
 - Internal (e.g., within organization control)
 - External (e.g., outside organization control)

- Third-party (e.g., outside of enterprise control)
- Location (e.g., on-premises, cloud, hybrid)

Domain 7: Security Operations

- **7.1 – Understand and comply with investigations**

- Evidence collection and handling
- Reporting and documentation
- Investigative techniques
- Digital forensics tools, tactics, and procedures
- Artifacts (e.g., data, computer, network, mobile device)

- **7.2 – Conduct logging and monitoring activities**

- Intrusion detection and prevention (IDPS)
- Security Information and Event Management (SIEM)
- Continuous monitoring and tuning
- Egress monitoring
- Log management
- Threat intelligence (e.g., threat feeds, threat hunting)
- User and Entity Behavior Analytics (UEBA)

- **7.3 – Perform Configuration Management (CM) (e.g., provisioning, baselining, automation)**

- **7.4 – Apply foundational security operations concepts**

- Need-to-know/least privilege
- Separation of Duties (SoD) and responsibilities
- Privileged account management
- Job rotation
- Service-level agreements (SLA)

- **7.5 – Apply resource protection**

- Media management
- Media protection techniques
- Data at rest/data in transit

- **7.6 – Conduct incident management**
 - Detection
 - Response
 - Mitigation
 - Reporting
 - Recovery
 - Remediation
 - Lessons learned
- **7.7 – Operate and maintain detection and preventative measures**
 - Firewalls (e.g., next generation, web application, network)
 - Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS)
 - Whitelisting/blacklisting
 - Third-party provided security services
 - Sandboxing
 - Honeypots/honeynets
 - Anti-malware
 - Machine learning and Artificial Intelligence (AI) based tools
- **7.8 – Implement and support patch and vulnerability management**
- **7.9 – Understand and participate in change management processes**
- **7.10 – Implement recovery strategies**
 - Backup storage strategies (e.g., cloud storage, onsite, offsite)
 - Recovery site strategies (e.g., cold vs. hot, resource capacity agreements)
 - Multiple processing sites
 - System resilience, high availability (HA), Quality of Service (QoS), and fault tolerance
- **7.11 – Implement Disaster Recovery (DR) processes**
 - Response
 - Personnel

- Communications (e.g., methods)
- Assessment
- Restoration
- Training and awareness
- Lessons learned
- **7.12 – Test Disaster Recovery Plans (DRP)**
 - Read-through/tabletop
 - Walkthrough
 - Simulation
 - Parallel
 - Full interruption
 - Communications (e.g., stakeholders, test status, regulators)
- **7.13 – Participate in Business Continuity (BC) planning and exercises**
- **7.14 – Implement and manage physical security**
 - Perimeter security controls
 - Internal security controls
- **7.15 – Address personnel safety and security concerns**
 - Travel
 - Security training and awareness (e.g., insider threat, social media impacts, two-factor authentication (2FA) fatigue)
 - Emergency management
 - Duress

Domain 8: Software Development Security

- **8.1 – Understand and integrate security in the Software Development Life Cycle (SDLC)**
 - Development methodologies (e.g., Agile, Waterfall, DevOps, DevSecOps, Scaled Agile Framework)
 - Maturity models (e.g., Capability Maturity Model (CMM), Software Assurance Maturity Model (SAMM))
 - Operation and maintenance

- Change management
- Integrated Product Team
- **8.2 – Identify and apply security controls in software development ecosystems**
 - Programming languages
 - Libraries
 - Tool sets
 - Integrated Development Environment
 - Runtime
 - Continuous Integration and Continuous Delivery (CI/CD)
 - Software configuration management (CM)
 - Code repositories
 - Application security testing (e.g., static application security testing (SAST), dynamic application security testing (DAST), software composition analysis, Interactive Application Security Test (IAST))
- **8.3 – Assess the effectiveness of software security**
 - Auditing and logging of changes
 - Risk analysis and mitigation
- **8.4 – Assess security impact of acquired software**
 - Commercial-off-the-shelf (COTS)
 - Open source
 - Third-party
 - Managed services (e.g., enterprise applications)
 - Cloud services (e.g., Software as a Service (SaaS), Infrastructure as a Service (IaaS), Platform as a Service (PaaS))
- **8.5 – Define and apply secure coding guidelines and standards**
 - Security weaknesses and vulnerabilities at the source-code level
 - Security of application programming interfaces (API)
 - Secure coding practices
 - Software-defined security

CCSP- Domain 1: Cloud Concepts, Architecture and Design (17%)

- **1.1 Understand Cloud Computing Concepts**
 - Cloud Computing Definitions
 - Cloud Computing Roles and Responsibilities
 - Essential Cloud Computing Characteristics
 - Building Block Technologies
- **1.2 Describe Cloud Reference Architecture**
 - Cloud Computing Activities
 - Cloud Service Capabilities
 - Cloud Service Categories
 - Cloud Deployment Models
 - Cloud Shared Considerations
 - Impact of Related Technologies
- **1.3 Understand Security Concepts Relevant to Cloud Computing**
 - Cryptography and Key Management
 - Identity and Access Control
 - Data and Media Sanitization
 - Network Security
 - Virtualization Security
 - Common Cloud Threats
 - Security Hygien
- **1.4 Understand Design Principles of Secure Cloud Computing**
 - Cloud Secure Data Lifecycle
 - Cloud-based Business Continuity (BC) and Disaster Recovery (DR) Planning

- Business Impact Analysis (BIA)
- Functional Security Requirements
- Security Considerations and Responsibilities for Different Cloud Categories
- Cloud Design Patterns
- DevOps Security
- **1.5 Evaluate Cloud Service Providers (CSP)**
 - Verification Against Criteria
 - System/Subsystem Product Certifications
- **1.6 Comprehend Artificial Intelligence (AI)/Machine Learning (ML)**
 - Cloud Threat Detection and Analysis
 - Data Source Validation and Verification
 - Security Orchestration, Automation and Response (SOAR)
 - Ethical Concerns
 - Regulatory Requirements

Domain 2: Cloud Data Security (20%)

- **2.1 Describe Cloud Data Concepts**
 - Cloud Data Lifecycle Phases
 - Data Dispersion
 - Data Flows
- **2.2 Design and Implement Cloud Data Storage Architectures**
 - Storage Types
 - Threats to Storage Types
- **2.3 Design and Apply Data Security Technologies and Strategies**
 - Encryption and Key Management
 - Hashing (e.g., data integrity, non-repudiation)
 - Data Obfuscation (e.g., masking, anonymization)
 - Tokenization
 - Data Loss Prevention (DLP)

- Keys, Secrets and Certificates Management
- **2.4 Implement Data Discovery**
 - Structured Data
 - Unstructured Data
 - Semi-Structured Data
 - Data Location
- **2.5 Plan and Implement Data Classification**
 - Data Classification Policies
 - Data Mapping
 - Data Labelling and Tagging
- **2.6 Design and Implement Information Rights Management (IRM)**
 - Objectives
 - Appropriate Tools
- **2.7 Plan and Implement Data Retention, Deletion and Archiving Policies**
 - Data Retention Policies
 - Data Deletion Procedures and Mechanisms
 - Data Archiving Procedures and Mechanisms
 - Legal Hold
- **2.8 Design and Implement Auditability, Traceability and Accountability of Data Events**
 - Definition of Event Sources and Requirement of Event Attributes
 - Logging, Storage and Analysis of Data Events
 - Chain of Custody and Non-repudiation
- **2.9 Comprehend Data Protection of AI and ML Data**
 - Data Set and Model Privacy
 - Data Set and Model Security

Domain 3: Cloud Platform and Infrastructure Security (17%)

- **3.1 Comprehend Cloud Infrastructure Components**
 - Physical Environment

- Network and Communications
- Compute
- Virtualization
- Storage
- Management Plane
- **3.2 Design a Secure Data Center**
 - Logical Design
 - Physical Design
 - Environmental Design
 - Design Resilience
- **3.3 Analyze Risks Associated with Cloud Infrastructure and Platforms**
 - Risk Assessment
 - Cloud Vulnerabilities, Threats and Attacks
 - Risk Treatment Strategies
- **3.4 Plan and Implementation of Security Controls**
 - Physical and Environmental Protection
 - System, Storage and Communication Protection
 - Identification, Authentication and Authorization in Cloud Environments
 - Audit Mechanisms
- **3.5 Plan Business Continuity (BC) and Disaster Recovery (DR)**
 - Business Continuity (BC)/Disaster Recovery (DR) Strategy
 - Business Requirements
 - Creation, Implementation and Testing of Plan

Domain 4: Cloud Application Security (16%)

- **4.1 Advocate Training and Awareness for Application Security**
 - Cloud Development Basics
 - Common Pitfalls
 - Common Cloud Vulnerabilities

- **4.2 Describe the Secure Software Development Life Cycle (SDLC) Proces**
 - Business Requirements
 - Phases and Methodologies (e.g., design, code, test, maintain, waterfall vs. agile)
- **4.3 Apply the Secure Software Development Life Cycle (SDLC)**
 - Cloud-Specific Risks
 - Threat Modelling
 - Avoid Common Vulnerabilities During Development
 - Secure Coding
 - Software Configuration Management (CM) and Versioning
- **4.4 Apply Cloud Software Assurance and Validation**
 - Functional and Non-functional Testing
 - Security Testing Methodologies
 - Quality Assurance (QA)
 - Abuse Case Testing
- **4.5 Use Verified Secure Software**
 - Securing Application Programming Interfaces (API)
 - Supply-Chain Management
 - Third-Party Software Management
 - Validated Open-Source Software
- **4.6 Comprehend and Apply the Specifics of Cloud Application Architecture**
 - Supplemental Security Components
 - Cryptography
 - Sandboxing
 - Application Virtualization and Orchestration
- **4.7 Design Appropriate Identity and Access Management (IAM) Solutions**
 - Federated Identity
 - Identity Providers (IdP)

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Cloud Access Security Broker (CASB)
- Secrets, Key, and Certificate Management

Domain 5: Cloud Security Operations (17%)

- **5.1 Build and Implement Physical and Logical Infrastructure for Cloud Environment**

- Hardware Specific Security Configuration Requirements
- Secure by Default
- Installation and Configuration of Management Plane Tools
- Virtual Hardware Specific Security Configuration Requirements
- Installation of Guest Operating System (OS) Virtualization Toolsets

- **5.2 Operate and Maintain Physical and Logical Infrastructure for Cloud Environment**

- Access Controls for Local and Remote Access
- Secure Network Configuration
- Network Security Controls
- Operating Systems Hardening through Application of Baselines, Monitoring and Remediation
- Patch Management
- Availability of Clustered Hosts
- Availability of Guest Operating System (OS)
- Performance and Capacity Monitoring
- Hardware Monitoring
- Configuration of Host and Guest OS Backup and Restore Functions
- Management Plane

- **5.3 Implement Operational Controls and Standards**

- Change Management
- Continuity Management
- Information Security Management

- Continual Service Improvement Management
- Incident Management
- Problem Management
- Release Management
- Deployment Management
- Configuration Management (CM)
- Service-Level Management
- Availability Management
- Capacity Management
- **5.4 Support Digital Forensics**
 - Forensic Data Collection Methodologies
 - Evidence Management
 - Collecting, Acquiring, and Preserving Digital Evidence
- **5.5 Manage Communication with Relevant Parties**
 - Vendors
 - Customers
 - Partners
 - Regulators
 - Other Stakeholders
- **5.6 Manage Security Operations**
 - Security Operations Center (SOC)
 - Intelligent Monitoring of Security Controls
 - Log Capture and Analysis
 - Incident Response (IR)
 - Vulnerability Assessments
 - Penetration Testing

Domain 6: Legal, Risk and Compliance (13%)

- **6.1 Articulate Legal Requirements and Unique Risks within the Cloud Environment**

- Conflicting International Legislation
- Evaluation of Legal Risks Specific to Cloud Computing
- Legal and Regulatory Frameworks and Guidelines
- eDiscovery
- Forensics Requirements
- **6.2 Understand Privacy Requirements**
 - Difference Between Contractual and Regulated Private Data
 - Country-Specific Legislation Related to Private Data
 - Jurisdictional Differences in Data Privacy
 - Standard Privacy Requirements
 - Privacy Impact Assessments (PIA)
- **6.3 Understand Audit Processes, Methodologies, and Required Adaptations for a Cloud Environment**
 - Internal and External Audit Controls
 - Impact of Audit Requirements
 - Identify Assurance Challenges of Virtualization and Cloud
 - Types of Audit Reports
 - Restrictions of Audit Scope Statements
 - Gap Analysis
 - Audit Planning
 - Internal Information Security Management System (ISMS)
 - Internal Information Security Controls System
 - Policies
 - Identification and Involvement of Relevant Stakeholders
 - Specialized Compliance Requirements for Highly Regulated Industries
 - Impact of Distributed Information Technology (IT) Model
- **6.4 Understand Implications of Cloud to Enterprise Risk Management**
 - Assess Providers Risk Management Programs
 - Difference Between Data Roles

- Regulatory Transparency Requirements
- Risk Treatment
- Different Risk Frameworks
- Metrics for Risk Management
- Assessment of Risk Environment
- **6.5 Understand Outsourcing and Cloud Contract Design**
 - Business Requirements
 - Vendor Management
 - Contract Management
 - Supply-Chain Management

SSCP- Domain 1. Security Operations and Administration (16%)

- 1.1 – Comply with codes of ethics
 - ISC2 Code of Ethics
 - Organizational code of ethics
- 1.2 – Understand security concepts
 - Confidentiality
 - Integrity
 - Availability
 - Accountability
 - Privacy
 - Non-repudiation
 - Least privilege
 - Segregation of Duties (SoD)
- 1.3 – Identify and implement security controls
 - Technical controls (e.g., session timeout, password aging)

- Physical controls (e.g., mantraps, cameras, locks)
- Administrative controls (e.g., security policies, standards, procedures, baselines)
- Assessing compliance
- Periodic audit and review
- 1.4 – Document and maintain functional security controls
 - Deterrent controls
 - Preventative controls
 - Detective controls
 - Corrective controls
 - Compensating controls
- 1.5 – Participate in asset management lifecycle (hardware, software and data)
 - Process, planning, design and initiation
 - Development/Acquisition
 - Inventory and licensing
 - Implementation/Assessment
 - Operation/Maintenance
 - Archiving and retention requirements
 - Disposal and destruction
- 1.6 – Participate in change management lifecycle
 - Change management (e.g., roles, responsibilities, processes)
 - Security impact analysis
 - Configuration management (CM)
- 1.7 – Participate in implementing security awareness and training (e.g., social engineering/phishing)
- 1.8 – Collaborate with physical security operations (e.g., data center assessment, badging)

Domain 2. Access Controls (15%)

- 2.1 – Implement and maintain authentication methods

- Single/Multi-Factor Authentication (MFA)
- Single Sign-On (SSO) (e.g., Active Directory Federation Services (ADFS), OpenID connect)
- Device authentication
- Federated access (e.g., Open Authorization 2 (OAuth2), Security Assertion Markup Language (SAML))
- 2.2 – Support internetwork trust architectures
 - Trust relationships (e.g., 1-way, 2-way, transitive, zero)
 - Internet, intranet and extranet
 - Third-party connections
- 2.3 – Participate in the identity management lifecycle
 - Authorization
 - Proofing
 - Provisioning/De-provisioning
 - Maintenance
 - Entitlement
 - Identity and Access Management (IAM) systems
- 2.4 – Understand and apply access controls
 - Mandatory
 - Discretionary
 - Role-based (e.g., attribute-, subject-, object-based)
 - Rule-based

Domain 3. Risk Identification, Monitoring, and Analysis (15%)

- 3.1 – Understand the risk management process
 - Risk visibility and reporting (e.g., risk register, Common Vulnerability Scoring (CVSS), sharing threat intelligence/Indicators of Compromise (IOC))
 - Risk management concepts (e.g., threat modeling, impact assessments)
 - Risk management frameworks

- Risk tolerance (e.g., appetite)
 - Risk treatment (e.g., acceptance, transference, mitigation, or avoidance)
- 3.2 – Understand legal and regulatory concerns (e.g., jurisdiction, limitations, privacy)
- 3.3 – Participate in security assessment and vulnerability management activities
 - Security testing
 - Risk review (e.g., internal, supplier, architecture)
 - Vulnerability management lifecycle
- 3.4 – Operate and monitor security platforms (e.g., continuous monitoring)
 - Source systems (e.g., applications, security appliances, network devices, and hosts)
 - Events of interest (e.g., anomalies, intrusions, unauthorized changes, compliance monitoring)
 - Log management
 - Event aggregation and correlation
- 3.5 – Analyze monitoring results
 - Security baselines and anomalies
 - Visualizations, metrics, and trends (e.g., notifications, dashboards, timelines)
 - Event data analysis
 - Document and communicate findings (e.g., escalation)

Domain 4. Incident Response and Recovery (14%)

- 4.1 – Support incident lifecycle, e.g., National Institute of Standards and Technology (NIST), International Organization for Standardization (ISO)
 - Preparation
 - Detection, analysis and escalation
 - Containment
 - Eradication

- Recovery
- Lessons learned/implementation of new countermeasure
- 4.2 – Understand and support forensic investigations
 - Legal (e.g., civil, criminal, administrative) and ethical principles
 - Evidence handling (e.g., first responder, triage, chain of custody, preservation of scene)
 - Reporting of analysis
- 4.3 – Understand and support Business Continuity Plan (BCP) and Disaster Recovery Plan (DRP)
 - Emergency response plans and procedures (e.g., pandemic, natural disaster, information system contingency, crisis management)
 - Interim or alternate processing strategies
 - Restoration planning
 - Backup and redundancy implementation
 - Testing and drills

Domain 5. Cryptography (9%)

- 5.1 – Understand cryptography
 - Confidentiality
 - Integrity and authenticity
 - Data sensitivity (e.g., Personally Identifiable Information (PII), Intellectual Property (IP), Protected Health Information (PHI))
 - Regulatory and industry best practices (e.g., International Organization for Standardization (ISO), Payment Card Industry Data Security Standards (PCI-DSS))
- 5.2 – Apply cryptography concepts
 - Hashing
 - Salting
 - Symmetric/Asymmetric encryption/Elliptic Curve Cryptography (ECC)
 - Non-repudiation (e.g., digital signatures/certificates, Hash-based Message Authentication Code (HMAC), audit trails)

- Strength of encryption algorithms and keys (e.g., Advanced Encryption Standards (AES), Rivest-Shamir-Adleman (RSA), 256-, 512-, 1024-, 2048-bit keys)
- Cryptographic attacks, cryptanalysis, and countermeasures (e.g., quantum computing)
- 5.3 – Understand and implement secure protocols
 - Services and protocols
 - Common use cases
 - Limitations and vulnerabilities
- 5.4 – Understand Public Key Infrastructure (PKI)
 - Fundamental key management concepts (e.g., storage, rotation, composition, generation, destruction, exchange, revocation, escrow)
 - Web of Trust (WOT)

Domain 6. Network and Communications Security (16%)

- 6.1 – Understand and apply fundamental concepts of networking
 - Open Systems Interconnection (OSI) and Transmission Control Protocol/Internet Protocol (TCP/IP) models
 - Network topologies
 - Network relationships (e.g., Peer-to-Peer (P2P), client server)
 - Transmission media types (e.g., wired, wireless)
 - Software-Defined Networking (SDN) (e.g., Software-Defined Wide Area Network (SD-WAN), network virtualization, automation)
 - Commonly used ports and protocols
- 6.2 – Understand network attacks
 - Distributed Denial of Service (DDoS)
 - Man-in-the-Middle (MITM)
 - Domain Name System (DNS) poisoning
 - Countermeasures (e.g., Content Delivery Networks (CDN))
- 6.3 – Manage network access controls

- Network access controls, standards and protocols (e.g., Remote Authentication Dial-In User Service (RADIUS), Institute of Electrical and Electronics Engineers (IEEE) 802.1X, Terminal Access Controller Access-Control System Plus (TACACS+))
- Remote access operation and configuration (e.g., thin client, Virtual Private Network (VPN))
- 6.4 – Manage network security
 - Logical and physical network devices placement of (e.g., passive, inline, virtual)
 - Segmentation (e.g., data/control plane, physical/logical, Virtual Local Area Network (VLAN), Access Control List (ACL), firewall zones, micro-segmentation)
 - Secure device management
- 6.5 – Operate and configure network-based security devices
 - Firewalls and proxies (e.g., filtering methods, Web Application Firewalls (WAF))
 - Network intrusion detection/prevention systems
 - Routers and switches
 - Traffic-shaping devices (e.g., Wide Area Network (WAN) optimization, load balancing)
- 6.6 – Secure wireless communications
 - Technologies (e.g., cellular network, Wi-Fi, Bluetooth, Near-Field Communication (NFC))
 - Authentication and encryption protocols (e.g., Wired Equivalent Privacy (WEP), Wi-Fi Protected Access (WPA), Extensible Authentication Protocol (EAP))
 - Internet of Things (IoT)

Domain 7. Systems and Application Security (15%)

- 7.1 – Identify and analyze malicious code and activity
 - Malware (e.g., spyware, scareware, ransomware, trojans, viruses, worms, rootkits, trapdoors, backdoors, fileless)
 - Malware countermeasures (e.g., scanners, anti-malware, code signing)

- Malicious activity (e.g., data theft, Distributed Denial of Service (DDoS), insider threat, botnet, zero-day exploits, web-based attacks, Advanced Persistent Threat (APT))
 - Malicious activity countermeasures (e.g., user awareness, patching, sandboxing, system hardening, isolation, Data Loss Prevention (DLP))
- 7.2 – Implement and operate endpoint device security
 - Host-based firewalls
 - Application whitelisting
 - Host-based Intrusion Prevention System (HIPS)
 - Endpoint encryption (e.g., whole disk encryption)
 - Trusted Platform Module (TPM)
 - Secure browsing
 - Endpoint Detection and Response (EDR)
- 7.3 – Administer Mobile Device Management (MDM)
 - Provisioning techniques (e.g., Corporate-Owned Personally Enabled (COPE), Bring Your Own Device (BYOD))
 - Containerization
 - Encryption
 - Mobile application management (MAM)
- 7.4 – Understand and configure cloud security
 - Deployment models (e.g., private, public, hybrid, community)
 - Service models (e.g., Platform as a Service (PaaS), Infrastructure as a Service (IaaS), and Software as a Service (SaaS))
 - Virtualization
 - Legal and regulatory concerns (e.g., privacy, jurisdiction, surveillance, data ownership, eDiscovery)
 - Data storage, processing, and transmission (e.g., recovery, resilience, archiving)
 - Third-party/outsourcing requirements (e.g., data portability, Service-Level Agreement (SLA), data destruction, auditing)
 - Shared responsibility model
- 7.5 – Operate and maintain secure virtual environments

- Containers
- Hypervisor
- Virtual appliances
- Continuity and resilience
- Attacks and countermeasures
- Shared storage

- **CC- Domain 1: Security Principles (26%)**

- 1.1: Understand the security concepts of information assurance
 - Confidentiality
 - Integrity
 - Availability
 - Authentication (e.g., methods of authentication, multi-factor authentication (MFA))
 - Non-repudiation
 - Privacy
- 1.2: Understand the risk management process
 - Risk management (e.g., risk priorities, risk tolerance)
 - Risk identification, assessment, and treatment
- 1.3: Understand security controls
 - Technical controls
 - Administrative controls
 - Physical controls
 - 1.4: Understand (ISC)² Code of Ethics
 - Professional code of conduct
- 1.4: Understand governance processes
 - Policies

- Procedures
 - Standards
 - Regulations and laws
- **Domain 2: Business Continuity (BC), Disaster Recovery (DR) & Incident Response Concepts (10%)**
 - 2.1: Understand business continuity (BC)
 - Purpose
 - Importance
 - Components
 - 2.2: Understand disaster recovery (DR)
 - Purpose
 - Importance
 - Components
 - 2.3: Understand incident response
 - Purpose
 - Importance
 - Components
- **Domain 3: Access Controls Concepts (22%)**
 - 3.1: Understand physical access controls
 - Physical security controls (e.g., badge systems, gate entry, environmental design)
 - Monitoring (e.g., security guards, closed-circuit television (CCTV), alarm systems, logs)
 - Authorized versus unauthorized personnel
 - 3.2: Understand logical access controls
 - Principle of least privilege
 - Segregation of duties
 - Discretionary access control (DAC)
 - Mandatory access control (MAC)
 - Role-based access control (RBAC)

- **Domain 4: Network Security (24%)**

- 4.1: Understand computer networking
 - Networks (e.g., Open Systems Interconnection (OSI) model, Transmission Control Protocol/Internet Protocol (TCP/IP) model, Internet Protocol version 4 (IPv4), Internet Protocol version 6 (IPv6), WiFi) Ports
 - Applications
- 4.2: Understand network threats and attacks
 - Types of threats (e.g., distributed denial-of-service (DDoS), virus, worm, Trojan, man-in-the-middle (MITM), side-channel)
 - Identification (e.g., intrusion detection system (IDS), host-based intrusion detection system (HIDS), network intrusion detection system (NIDS))
 - Prevention (e.g., antivirus, scans, firewalls, intrusion prevention system (IPS))
- 4.3: Understand network security infrastructure
 - On-premises (e.g., power, data center/closets, Heating, Ventilation, and Air Conditioning (HVAC), environmental, fire suppression, redundancy, memorandum of understanding (MOU)/memorandum of agreement (MOA))
 - Design (e.g., network segmentation (demilitarized zone (DMZ), virtual local area network (VLAN), virtual private network (VPN), micro-segmentation), defense in depth, Network Access Control (NAC) (segmentation for embedded systems, Internet of Things (IoT))
 - Cloud (e.g., service-level agreement (SLA), managed service provider (MSP), Software as a Service (SaaS), Infrastructure as a Service (IaaS), Platform as a Service (PaaS), hybrid)

- **Domain 5: Security Operations (18%)**

- 5.1: Understand data security
 - Encryption (e.g., symmetric, asymmetric, hashing)
 - Data handling (e.g., destruction, retention, classification, labeling)
 - Logging and monitoring security events
- 5.2: Understand system hardening

- Configuration management (e.g., baselines, updates, patches)
- 5.3: Understand best practice security policies
 - Data handling policy
 - Password policy
 - Acceptable Use Policy (AUP)
 - Bring your own device (BYOD) policy
 - Change management policy (e.g., documentation, approval, rollback)
 - Privacy policy
- 5.4: Understand security awareness training
 - Purpose/concepts (e.g., social engineering, password protection)
 - Importance

CISA- CISA Domain 1: Information System Auditing Process (18%)

- A-Planning
 - IS Audit Standards, Guidelines, and Codes of Ethics
 - Business Processes
 - Types of Controls
 - Risk-Based Audit Planning
 - Types of Audits and Assessments
- B-Execution
 - Audit Project Management
 - Sampling Methodology
 - Audit Evidence Collection Techniques
 - Data Analytics
 - Reporting and Communication Techniques
 - Quality Assurance and Improvement of the Audit Process

CISA Domain 2: Governance and Management of IT (18%)

- A-IT Governance
 - Laws, Regulations, and Industry Standards
 - Organizational Structure, IT Governance, and IT Strategy
 - IT Policies, Standards, Procedures, and Practices
 - Enterprise Architecture and Considerations
 - Enterprise Risk Management
 - Privacy Program and Principles
 - Data Governance and Classification
- B-IT Management
 - IT Resource Management
 - IT Vendor Management
 - IT Performance Monitoring and Reporting
 - Quality Assurance and Quality Management of IT

CISA Domain 3: Information Systems Acquisition, Development, and Implementation (12%)

- A-Information Systems Acquisition and Development
 - Project Governance and Management
 - Business Case and Feasibility Analysis
 - System Development Methodologies
 - Control Identification and Design
- B-Information Systems Implementation
 - System Readiness and Implementation Testing
 - Implementation, Configuration, and Release Management
 - System Migration, Infrastructure Deployment, and Data Conversion
 - Post-implementation Review

CISA Domain 4: Information Systems Operations and Business Resilience (26%)

- A-Information Systems Operations
 - IT Components
 - IT Asset Management
 - Job Scheduling and Production Process Automation
 - System Interfaces
 - Shadow IT and End-User Computing
 - Systems Availability and Capacity Management
 - Problem and Incident Management
 - IT Change, Configuration, and Patch Management
 - Operational Log Management
 - IT Service Level Management
 - Database Management
- B-Business Resilience
 - Business Impact Analysis
 - System and Operational Resilience
 - Data Backup, Storage, and Restoration
 - Business Continuity Plan
 - Disaster Recovery Plans

CISA Domain 5: Protection of Information Assets (26%)

- A- Information Asset Security and Controls
 - Information Asset Security Frameworks, Standards, and Guidelines
 - Physical and Environmental Controls
 - Identity and Access Management
 - Network and End-Point Security
 - Data Loss Prevention
 - Data Encryption
 - Public Key Infrastructure
 - Cloud and Virtualized Environments

- Mobile, Wireless, and Internet-of-Things Devices
- **B- Security Event Management**
 - Security Awareness Training and Programs
 - Information System Attack Methods and Techniques
 - Security Testing Tools and Techniques
 - Security Monitoring Tools and Techniques
 - Security Incident Response Management
 - Evidence Collection and Forensics

CISM- Domain 1 – Information Security Governance – 17%

- **A–Enterprise Governance**
 - Organizational Culture
 - Legal, Regulatory, and Contractual Requirements
 - Organizational Structures, Roles, and Responsibilities
- **B–Information Security Strategy**
 - Information Security Strategy Development
 - Information Governance Frameworks and Standards
 - Strategic Planning (e.g., Budgets, Resources, Business Case)

Domain 2 – Information Security Risk Management – 20%

- **A–Information Security Risk Assessment**
 - Emerging Risk and Threat Landscape
 - Vulnerability and Control Deficiency Analysis
 - Risk Assessment and Analysis
- **B–Information Security Risk Response**
 - Risk Treatment / Risk Response Options
 - Risk and Control Ownership

- Risk Monitoring and Reporting

Domain 3 – Information Security Program – 33%

- **A–Information Security Program Development**

- Information Security Program Resources (e.g., People, Tools, Technologies)
- Information Asset Identification and Classification
- Industry Standards and Frameworks for Information Security
- Information Security Policies, Procedures, and Guidelines
- Information Security Program Metrics

- **B–Information Security Program Management**

- Information Security Control Design and Selection
- Information Security Control Implementation and Integrations
- Information Security Control Testing and Evaluation
- Information Security Awareness and Training
- Management of External Services (e.g., Providers, Suppliers, Third Parties, Fourth Parties)
- Information Security Program Communications and Reporting

Domain 4 – Incident Management – 30%

- **A–Incident Management Readiness**

- Incident Response Plan
- Business Impact Analysis (BIA)
- Business Continuity Plan (BCP)
- Disaster Recovery Plan (DRP)
- Incident Classification/Categorization
- Incident Management Training, Testing, and Evaluation

- **B–Incident Management Operations**

- Incident Management Tools and Techniques
- Incident Investigation and Evaluation

- Incident Containment Methods
- Incident Response Communications (e.g., Reporting, Notification, Escalation)
- Incident Eradication and Recovery
- Post-Incident Review Practices

- **CRISC- Domain 1: Governance 26%**
 - **Part A: Organizational Governance**
 - Strategy, Goals, and Objectives
 - Organizational Structure, Roles, and Responsibilities
 - Organizational Culture and Ethics
 - Policies and Standards
 - Business Processes and Resilience
 - Organizational Asset Management
 - **Part B: Risk Governance**
 - Enterprise Risk Management
 - Lines of Defense
 - Risk Profile
 - Risk Appetite and Risk Tolerance
 - Risk Frameworks, Legal, Regulatory, and Contractual Requirements

- **Domain 2: IT Risk Assessment 22%**
 - **Part A: Risk Identification**
 - Risk Events
 - Threat Modeling and Threat Landscape
 - Vulnerability Management
 - Risk Scenario Development and Evaluation

- **Part B: Risk Analysis**
 - Risk Assessment Concepts and Standards
 - Business Impact Analysis (BIA)
 - Risk Register
 - Risk Analysis Methodologies
 - Inherent, Residual, and Current Risk
- **Domain 3: Risk Response and Reporting 32%**
 - **Part A: Risk Response**
 - Risk Response Options
 - Risk and Control Ownership
 - Vendor/Supply Chain Risk Management
 - Issues, Findings, Exceptions, and Exemptions Management
 - **Part B: Control Design and Implementation**
 - Control Frameworks, Types, and Standards
 - Control Design, Selection, Implementation, and Analysis
 - Control Testing Methodologies
 - **Part C: Risk Monitoring and Reporting**
 - Risk Action Plans
 - Data Collection, Aggregation, Analysis, and Validation
 - Risk and Control Metrics
 - Risk and Control Monitoring and Reporting Techniques
 - Monitoring and Reporting of Emerging Risks
- **Domain 4 : Information Technology and Security 20%**
 - **Part A: Information Technology Principles**
 - Technology Roadmaps and Enterprise Architecture (EA)
 - Operations Management
 - System Development Life Cycle (SDLC)
 - Data Lifecycle Management
 - Portfolio and Project Management

- Technology Resilience and Disaster Response/Recovery
 - Emerging Technologies
 - **Part B: Information Security Principles**
 - Security Concepts, Frameworks, and Standards
 - Security/Risk Awareness and Training
 - Data Privacy and Data Protection Principles
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- CGEIT- **Domain 1:** Framework for the Governance of Enterprise IT.
 - **Domain 2:** Strategic Management.
 - **Domain 3:** Benefits Realization.
 - **Domain 4:** Risk Optimization.
 - **Domain 5:** Resource Optimization.

IAPP-CIPP/E

- **Domain 1: Introduction to European Data Protection**
 - Understand the origins and historical context of European data protection laws.
 - Understand the roles and functions of significant European Union institutions.
 - Understand the legislative framework underpinning the principles of European data protection.
- **Domain 2: European Data Protection Law and Regulation**
 - Understand basic GDPR data protection concepts.
 - Understand the requirements involved in maintaining the security of personal data.

- Understand data subjects' rights.
- **Domain 3: European Data Processing**
 - Understand the principles of European data processing
 - Know what constitutes a lawful processing basis.
 - Understand information provision obligations.
 - Understand the principles of, and the risks involved in, international data transfers.
- **Domain 4: European Data Protection: Scope and Accountability**
 - Understand issues related to the territorial and material scope of the GDPR.
 - Understand the various accountability requirements under the GDPR.
 - Understand the European data protection supervision and enforcement structure.
 - Understand the consequences for GDPR violations.
- **Domain 5: Compliance with European Data Protection Law and Regulation**
 - Understand how to comply with European data protection laws and regulations in the workplace, specifically as they relate to employment relationships.
 - Understand how to comply with European data protection laws and regulations related to surveillance activities.
 - Understand how to comply with European data protection laws and regulations related to direct marketing.
 - Understand how to comply with European data protection laws and regulations related to internet technology and communications.
- **IAPP-CIPM- Module 1: Introduction to privacy program management**Identifies privacy program management responsibilities and describes the role of accountability in privacy program management.

- **Module 2: Privacy governance**Examines considerations for developing and implementing a privacy program, including the position of the privacy function within the organization, role of the DPO, program scope and charter, privacy strategy, support and ongoing involvement of key functions and privacy frameworks.
- **Module 3: Applicable laws and regulations**Discusses the regulatory environment, common elements across jurisdictions and strategies for aligning compliance with organizational strategy.
- **Module 4: Data assessments**Relates practical processes for creating and using data inventories/maps, gap analysis, privacy assessments, privacy impact assessments/data protection impact assessments and vendor assessments.
- **Module 5: Policies**Describes common types of privacy-related policies, outlines components and offers strategies for implementation.
- **Module 6: Data subject rights**Discusses operational considerations for communicating and ensuring data subject rights, including privacy notice, choice and consent, access and rectification, data portability, and erasure.
- **Module 7: Training and awareness**Outlines strategies for developing and implementing privacy training and awareness programs.
- **Module 8: Protecting personal information**Examines a holistic approach to protecting personal information through privacy by design.
- **Module 9: Data breach incident plans**Provides guidance on planning for and responding to a data security incident or breach.
- **Module 10: Monitoring and auditing program performance**Relates common practices for monitoring, measuring, analyzing and auditing privacy program performance.

Network +

Chapter 1: Fundamentals

- Module A: Networking concepts
- Module B: Classifying networks
- Module C: Network models
- Module D: The troubleshooting process

Chapter 2: Physical networks

- Module A: Connection technologies
- Module B: Network devices
- Module C: Copper media
- Module D: Optical media
- Module E: Ethernet standards

Chapter 3: TCP/IP networks

- Module A: IP addressing
- Module B: Core protocols
- Module C: Network ports and applications

Chapter 4: Internetworking

- Module A: Switching
- Module B: Routing

Chapter 5: Wireless LANs

- Module A: Wireless networks
- Module B: Wireless LAN standards

Chapter 6: Wide area networks

- Module A: Internet connections
- Module B: WAN infrastructure

Chapter 7: Cybersecurity principles

- Module A: Goals and threats
- Module B: Digital security
- Module C: Transport encryption

Chapter 8: Defending networks

- Module A: Network security components
- Module B: Network authentication systems
- Module C: Hardening networks

Chapter 9: Evolving network technologies

- Module A: Network convergence

- Module B: Virtual and cloud systems

Chapter 10: Network operations

- Module A: Monitoring and optimization
- Module B: Fault tolerance and disaster recovery
- Module C: Incident response

Chapter 11: Network planning

- Module A: Network policy design
- Module B: Network installation
- Module C: Maintenance and upgrades

Pentest+

1.0 planning and scoping

- 1.1 Compare and contrast governance, risk, and compliance concepts
- 1.2 Explain the importance of scoping organizations/customer requirements
- 1.3 Given a scenario, demonstrate an ethical hacking mindset by maintaining professionalism and integrity.

2.0 Information gathering and vulnerability scanning

- 2.1 Given a scenario, perform passive reconnaissance
- 2.2 Given a scenario, perform active reconnaissance
- 2.3 Given a scenario, analyze the results of a reconnaissance exercise
- 2.4 Given a scenario, perform vulnerability scanning

3.0 Attacks and Exploits

- 3.1 Given a scenario, research attack vectors and perform network attacks
- 3.2 Given a scenario, research attack vectors and perform wireless attacks
- 3.3 Given a scenario, research attack vectors and perform application-based attacks

- 3.4 Given a scenario, research attack vectors and perform attacks on cloud technologies
- 3.5 Explain common attacks and vulnerabilities against specialized systems
- 3.6 Given a scenario, perform a social engineering or physical attack
- 3.7 Given a scenario, perform post-exploitation techniques

4.0 Reporting and Communication

- 4.1 Compare and contrast important components of written reports
- 4.2 Given a scenario, analyze the findings and recommend the appropriate remediation within a report
- 4.3 Explain the importance of communication during the penetration testing process
- 4.4 Explain post-report delivery activities

5.0 Tools and Code Analysis

- 5.1 Explain the basic concepts of scripting and software development
- 5.2 Given a scenario, analyze a script or code sample for use in a penetration test
- 5.3 Explain use cases of the following tools during the phases of a penetration test

CPENT- CPENT Certification Training Course consists of 14 modules and tests the abilities of a penetration tester in almost all the vectors of cybersecurity, some of which have been introduced for the first time in any penetration certification

- **Module 01:** Introduction to Penetration Testing
- **Module 02:** Penetration Testing Scoping and Engagement
- **Module 03:** Open Source Intelligence (OSINT)
- **Module 04:** Social Engineering Penetration Testing
- **Module 05:** Network Penetration Testing – External
- **Module 06:** Network Penetration Testing– Internal

- **Module 07:** Network Penetration Testing – Perimeter Devices
- **Module 08:** Web Application Penetration Testing
- **Module 09:** Wireless Penetration Testing
- **Module 10:** IoT Penetration Testing
- **Module 11:** OT/SCADA Penetration Testing
- **Module 12:** Cloud Penetration Testing
- **Module 13:** Binary Analysis and Exploitation
- **Module 14:** Report Writing and Post Testing Actions

CSA CCSK-

Domain 1: Cloud Computing Concepts & Architectures

- Introduction
- Learning Objectives
- 1.1 Defining Cloud Computing
 - 1.1.1 Abstraction & Orchestration
- 1.2 Cloud Computing Models
 - 1.2.1 Essential Characteristics
 - 1.2.2 Cloud Service Models
 - 1.2.2.1 Infrastructure as a Service (IaaS)
 - 1.2.2.2 Platform as a Service (PaaS)
 - 1.2.2.3 Software as a Service (SaaS)
 - 1.2.3 Cloud Deployment Models
 - 1.2.4 CSA Enterprise Architecture Model
- 1.3 Cloud Security Scope, Responsibilities, & Models

- 1.3.1 Shared Security Responsibility Mode

Domain 2: Cloud Governance

- Introduction
- Learning Objectives
- 2.1. Cloud Governance
- 2.2 The Governance Hierarchy
 - 2.2.1 Cloud Security Frameworks
 - 2.2.2 Policies

Domain 3: Risk, Audit, & Compliance

- Introduction
- Learning Objectives
- 3.1. Cloud Risk Management
 - 3.1.1 Cloud Risks
 - 3.1.2 Understanding Cloud Risk Management
 - 3.1.3 Assessing Cloud Services
 - 3.1.4 The Cloud Register
- 3.2 Compliance & Audit
 - 3.2.1 Jurisdictions
 - 3.2.2 Cloud-Relevant Laws & Regulations Examples
 - 3.2.3 Compliance Inheritance
 - 3.2.4 Artifacts of Compliance
- 3.3 Governance, Risk, Compliance Tools & Technologies

Domain 4: Organization Management

- Introduction
- Learning Objectives
- 4.1 Organization Hierarchy Models

- 4.1.1 Definitions
- 4.1.2 Organization Capabilities Within a Cloud Service Provider
- 4.1.3 Building a Hierarchy Within a Provider
- 4.2 Managing Organization-Level Security Within a Provider
 - 4.2.1 Identity Provider & User/Group/Role Mappings
 - 4.2.2 Common Organization Shared Services
- 4.3 Considerations for Hybrid & Multi-Cloud Deployments
 - 4.3.1 Organization Management for Hybrid Cloud Security
 - 4.3.2 Organization Management for Multi-Cloud Security
 - 4.3.3 Organization Management for SaaS Hybrid & Multi-Cloud

Domain 5: Identity & Access Management

- Introduction
- Learning Objectives
- 5.1 Fundamental Terms
- 5.2 Federation
 - 5.2.1 Common Federation Standards
 - 5.2.2 How Federated Identity Management Works
 - 5.2.3 Managing Users & Identities for Cloud Computing
- 5.3 Strong Authentication & Authorization
 - 5.3.1 Authentication & Credentials
 - 5.3.2 Entitlement & Access Management
 - 5.3.3 Privileged User Management

Domain 6: Security Monitoring

- Introduction
- Learning Objectives
- 6.1 Cloud Monitoring
 - 6.1.1 Logs & Events

- 6.2 Beyond Logs – Posture Management
- 6.3 Cloud Telemetry Sources
 - 6.3.1 Management Plane Logs
 - 6.3.2 Service & Application Logs
 - 6.3.3 Resource Logs
 - 6.3.4 Cloud Native Tools
- 6.4 Collection Architectures
 - 6.4.1 Log Storage & Retention
 - 6.4.2 Cascading Log Architecture
- 6.5 AI for Security Monitoring

Domain 7: Infrastructure & Networking

- Introduction
- Learning Objectives
- 7.1 Cloud Infrastructure Security
 - 7.1.1 Foundational Infrastructure Security Techniques
 - 7.1.2 CSP Infrastructure Security Responsibilities
 - 7.1.3 Infrastructure Resilience
- 7.2 Cloud Network Fundamentals
 - 7.2.1 Cloud Networks are Software-Defined Networks
 - 7.2.2 Cloud Connectivity
- 7.3 Cloud Network Security & Secure Architecture
 - 7.3.1 Preventative Security Measures
 - 7.3.2 Detective Security Measures
- 7.4 Infrastructure as Code (IaC)
- 7.5 Zero Trust for Cloud Infrastructure & Networks
 - 7.5.1 Software-Defined Perimeter & ZT Network Access
- 7.6 Secure Access Service Edge (SASE)

Domain 8: Cloud Workload Security

- Introduction
- Learning Objectives
- 8.1 Introduction to Cloud Workload Security
 - 8.1.1 Types of Cloud Workloads
 - 8.1.2 Impact on Workload Security Controls
- 8.2 Securing Virtual Machines
 - 8.2.1 Virtual Machine Challenges & Mitigations
 - 8.2.2 Creating Secure VM Images with Factories
 - 8.2.3 Snapshots & Public Exposures/Exfiltration
- 8.3 Securing Containers
 - 8.3.1 Container Image Creation
 - 8.3.2 Container Networking
 - 8.3.3 Container Orchestration & Management Systems
 - 8.3.4 Container Orchestration Security
 - 8.3.5 Runtime Protection for Containers
- 8.4 Securing Serverless and Function as a Service
 - 8.4.1 FaaS Security Issues
 - 8.4.2 IAM for Serverless
 - 8.4.3 Environment Variables & Secrets
- 8.5 Securing AI Workloads
 - 8.5.1 AI-System Threats
 - 8.5.2 AI Risk Mitigation and Shared Responsibilities

Domain 9: Data Security

- Introduction
- Learning Objectives
- 9.1 Primer on Cloud Storage
 - 9.1.1 Volume/Block Storage

- 9.1.2 Object Storage
 - 9.1.3 Database Storage
 - 9.1.4 Other Types of Storage
- 9.2 Data Security Tools and Techniques
 - 9.2.1 Data Classification
 - 9.2.2 Identity and Access Management
 - 9.2.3 Access Policies
 - 9.2.4 Encryption and Key Management
 - 9.2.5 Data Loss Prevention
- 9.3 Cloud Data Encryption at Rest
 - 9.3.1 Cloud Data Key Management Strategies
 - 9.3.2 Data Encryption Recommendations
- 9.4 Data Security Posture Management
- 9.5 Object Storage Security
- 9.6 Data Security for Artificial Intelligence
 - 9.6.1 AI as a Service

Domain 10: Application Security

- Introduction
- Learning Objectives
- 10.1 Secure Development Lifecycle
 - 10.1.1 SDLC Stages
 - 10.1.2 Threat Modeling
 - 10.1.3 Testing: Pre-Deployment
 - 10.1.4 Testing: Post Deployment
- 10.2 Architecture's Role in Secure Cloud Applications
 - 10.2.1 Cloud Impacts on Architecture-Level Security
 - 10.2.2 Architectural Resilience
- 10.3 Identity & Access Management and Application Security

- 10.3.1 Secrets Management
- 10.4 Dev Ops & DevSecOps

Domain 11: Incident Response & Resilience

- Introduction
- Learning Objectives
- 11.1 Incident Response
 - 11.1.1 Incident Response Lifecycle
- 11.2 Preparation
 - 11.2.1 Incident Response Preparation & Cloud Service Providers
 - 11.2.2 Training for Cloud Incident Responders
- 11.3 Detection & Analysis
 - 11.3.1 Cloud Impact on Incident Response Analysis
 - 11.3.2 Cloud System Forensics
- 11.4 Containment, Eradication, & Recovery
 - 11.4.1 Containment
 - 11.4.2 Eradication
 - 11.4.3 Recovery
- 11.5 Post Incident Analysis

Domain 12: Related Technologies & Strategies

- Introduction
- Learning Objectives
- 12.1 Zero Trust
 - 12.1.1 Technical Objectives of Zero Trust
 - 12.1.2 Zero Trust Pillars & Maturity Model
 - 12.1.3 Zero Trust & Cloud Security
- 12.2 Artificial Intelligence
 - 12.2.1 Characteristics of AI Workloads

- Next Steps.

CDPSE= **CDPSE Course Outline**

Upon completing this course, you will be able to:

- Identify the privacy program requirements
- Describe the types of privacy protection legal models
- Identify the common privacy laws and regulations
- Identify the various privacy standards
- Identify the different types of documentation necessary for data privacy management
- Explain the requirements to address the data subjects' right
- Identify roles & responsibilities of various stakeholders
- Define & execute a privacy awareness program
- Apply privacy controls for the privacy risks and issues
- Define privacy incident management program
- Identify the common privacy-related vulnerabilities caused by the problematic data actions
- Identify the methods of exploiting these vulnerabilities leading to privacy breaches and harms
- Identify the common established PIA methodologies
- Describe the NIST privacy risk assessment methodology and EU GDPR
- Describe Data Protection Impact Assessment (DPIA)
- Identify various types of computing infrastructure
- Identify the advantages and limitations of cloud computing
- Recognize the responsibilities of the Cloud Service Provider and the cloud consumer in a shared responsibility model
- Describe Secure Development Life Cycle
- List the various considerations for endpoint security

- Describe the elements and principles of privacy by design
- Identify the best practices for system hardening
- Explain the steps involved in application and system hardening to protect the enterprise's software/applications from privacy breach
- Describe the privacy considerations required for applications using APIs and web services
- Recognize the risks associated with the various communication protocols
- Explain the steps to create a data inventory & classification
- Describe the four process areas of data quality
- Illustrate the different data flow diagrams
- Explain data analytics and its privacy concerns
- Explain data minimalization
- Explain data migration & storage requirements
- Explain data warehousing
- Explain data retention, archiving & destruction

- ISO 27001 Foundations- Introduction to the ISO/IEC 27000 family of standards
- Introduction to management systems and the process approach
- Fundamental principles information security
- General requirements: presentation of the clauses 4 to 8 of ISO/IEC 27001
- Implementation phases of ISO/IEC 27001 framework
- Continual improvement of Information Security
- Conducting an ISO/IEC 27001 certification audit
- Principles and design of information security controls
- Documentation of an information security control environment
- Monitoring and reviewing the information security controls

- Examples of implementation of information security controls based on ISO 27002 best practices

ISO 27001 Lead Implementer

Introduction to ISO/IEC 27001 and initiation of an ISMS

Section 1: Training course objectives and structure

- Introduction
- General information
- Learning objectives
- Educational approach
- Examination and certification

Section 2: Standards and regulatory frameworks

- What is ISO?
- The ISO/IEC 27000 family of standards
- Advantages of ISO/IEC 27001

Section 3: Information Security Management System (ISMS)

- Definition of a management system
- Management system standards
- Integrated management systems
- Definition of an ISMS
- Process approach
- Overview — Clauses 4 to 10
- Overview — Annex A

Section 4: Fundamental information security concepts and principles

- Information and asset
- Information security

- Availability, confidentiality, and integrity
- Vulnerability, threat, and impact
- Information security risk
- Classification of security controls

Section 5: Initiation of the ISMS implementation

- Define the approach to the ISMS implementation
- Proposed implementation approaches
- Application of the proposed implementation approaches
- Choose a methodological framework to manage the implementation of an ISMS
- Approach and methodology
- Alignment with best practices

Section 6: Understanding the organization and its context

- Mission, objectives, values, and strategies of the organization
- ISMS objectives
- Preliminary scope definition
- Internal and external environment
- Key processes and activities
- Interested parties
- Business requirements

Section 7: ISMS scope

- Boundary of the ISMS
- Organizational boundaries
- Information security boundaries
- Physical boundaries
- ISMS scope statement

Planning the implementation of an ISMS

Section 8: Leadership and project approval

- Business case

- Resource requirements
- ISMS project plan
- ISMS project team
- Management approval

Section 9: Organizational structure

- Organizational structure
- Information security coordinator
- Roles and responsibilities of interested parties
- Roles and responsibilities of key committees

Section 10: Analysis of the existing system

- Determine the current state
- Conduct the gap analysis
- Establish maturity targets
- Publish a gap analysis report

Section 11: Information security policy

- Types of policies
- Policy models
- Information security policy
- Specific security policies
- Management policy approval
- Publication and dissemination
- Training and awareness sessions
- Control, evaluation, and review

Section 12: Risk management

- ISO/IEC 27005
- Risk assessment approach
- Risk assessment methodology
- Risk identification
- Risk estimation

- Risk evaluation
- Risk treatment
- Residual risk

Section 13: Statement of Applicability

- Drafting the Statement of Applicability
- Management approval
- Review and selection of the applicable information security controls
- Justification of selected controls
- Justification of excluded controls

Implementation of an ISMS

Section 14: Documented information management

- Value and types of documented information
- Master list of documented information
- Creation of templates
- Documented information management process
- Implementation of a documented information management system
- Management of records

Section 15: Selection and design of controls

- Organization's security architecture
- Preparation for the implementation of controls
- Design and description of controls

Section 16: Implementation of controls

- Implementation of security processes and controls
- Introduction of Annex A controls

Section 17: Trends and technologies

- Big data
- The three V's of big data
- Artificial intelligence
- Machine learning

- Cloud computing
- Outsourced operations
- The impact of new technologies in information security

Section 18: Communication

- Principles of an efficient communication strategy
- Information security communication process
- Establishing communication objectives
- Identifying interested parties
- Planning communication activities
- Performing a communication activity
- Evaluating communication

Section 19: Competence and awareness

- Competence and people development
- Difference between training, awareness, and communication
- Determine competence needs
- Plan the competence development activities
- Define the competence development program type and structure
- Training and awareness programs
- Provide the trainings
- Evaluate the outcome of trainings

Section 20: Security operations management

- Change management planning
- Management of operations
- Resource management
- ISO/IEC 27035-1 and ISO/IEC 27035-2
- ISO/IEC 27032
- Information security incident management policy
- Process and procedure for incident management
- Incident response team

- Incident management security controls
- Forensics process
- Records of information security incidents
- Measure and review of the incident management process

ISMS monitoring, continual improvement, and preparation for the certification audit

Section 21: Monitoring, measurement, analysis, and evaluation

- Determine measurement objectives
- Define what needs to be monitored and measured
- Establish ISMS performance indicators
- Report the results

Section 22: Internal audit

- What is an audit?
- Types of audits
- Create an internal audit program
- Designate a responsible person
- Establish independence, objectivity, and impartiality
- Plan audit activities
- Perform audit activities
- Follow up on nonconformities

Section 23: Management review

- Preparing a management review
- Conducting a management review
- Management review outputs
- Management review follow-up activities

Section 24: Treatment of nonconformities

- Root-cause analysis process
- Root-cause analysis tools
- Corrective action procedure

- Preventive action procedure

Section 25: Continual improvement

- Continual monitoring process
- Maintenance and improvement of the ISMS
- Continual update of the documented information
- Documentation of the improvements

Section 26: Preparing for the certification audit

- Selecting the certification body
- Preparing for the certification audit
- Stage 1 audit
- Stage 2 audit
- Follow-up audit
- Certification decision

Section 27: Practical Approach of ISMS Implementation

- Planning for ISMS Implementation
- Gap Assessment
- Risk Assessment
- Risk Treatment
- Creating Statement of Applicability
- Internal Audit Process
- Management Review
- Documentation

Section 28: Preparation for Exam and Interview

Module 1: Introduction to COBIT 2019

Topics Covered

- Overview of IT Governance and Management
- Evolution of COBIT Framework
- COBIT 2019 vs COBIT 5
- Benefits of COBIT Implementation
- Key Terminologies and Concepts
- Governance vs Management
- Enterprise Governance of Information and Technology (EGIT)

Learning Outcomes

- Understand the purpose and scope of COBIT 2019
 - Differentiate governance from management
 - Explain the role of COBIT in enterprise IT governance
-

Module 2: COBIT 2019 Framework Principles

Topics Covered

- Governance System Principles
 - Provide Stakeholder Value
 - Holistic Approach
 - Dynamic Governance System
 - Distinct Governance from Management
 - Tailored to Enterprise Needs
 - End-to-End Governance System
- Governance Framework Principles
 - Conceptual Model
 - Open and Flexible
 - Aligned to Major Standards

Learning Outcomes

- Understand the six governance system principles

- Explain governance framework principles
 - Apply principles to governance scenarios
-

Module 3: Governance and Management Objectives

Topics Covered

- Introduction to Governance Objectives
- Introduction to Management Objectives
- COBIT Core Model
- Governance Domain:
 - EDM (Evaluate, Direct and Monitor)
- Management Domains:
 - APO (Align, Plan and Organize)
 - BAI (Build, Acquire and Implement)
 - DSS (Deliver, Service and Support)
 - MEA (Monitor, Evaluate and Assess)

Learning Outcomes

- Understand the purpose of governance and management objectives
 - Identify responsibilities within each domain
-

Module 4: COBIT Governance System Components

Topics Covered

- Processes
- Organizational Structures
- Policies and Procedures
- Information Flows and Items
- Culture, Ethics and Behavior
- People, Skills and Competencies
- Services, Infrastructure and Applications

Learning Outcomes

- Explain governance system components
 - Understand how components interact to support governance
-

Module 5: Governance and Management Domains Deep Dive

Topics Covered

EDM Domain

- EDM01 Governance Framework Setting and Maintenance
- EDM02 Benefits Delivery
- EDM03 Risk Optimization
- EDM04 Resource Optimization
- EDM05 Stakeholder Transparency

APO Domain

- APO01 Managed I&T Management Framework
- APO02 Managed Strategy
- APO03 Managed Enterprise Architecture
- APO12 Managed Risk
- APO13 Managed Security

Learning Outcomes

- Understand domain objectives and activities
 - Map governance requirements to enterprise goals
-

Module 6: Build, Acquire and Implement (BAI)

Topics Covered

- BAI01 Managed Programs
- BAI02 Managed Requirements
- BAI03 Managed Solutions Identification
- BAI05 Organizational Change Enablement
- BAI06 Managed IT Changes
- BAI09 Managed Assets

- BAI11 Managed Projects

Learning Outcomes

- Understand project governance and implementation processes
 - Manage enterprise IT initiatives using COBIT practices
-

Module 7: Deliver, Service and Support (DSS)

Topics Covered

- DSS01 Managed Operations
- DSS02 Managed Service Requests and Incidents
- DSS03 Managed Problems
- DSS04 Managed Continuity
- DSS05 Managed Security Services
- DSS06 Managed Business Process Controls

Learning Outcomes

- Explain service management concepts
 - Understand operational governance and support functions
-

Module 8: Monitor, Evaluate and Assess (MEA)

Topics Covered

- MEA01 Performance and Conformance
- MEA02 Internal Control System
- MEA03 Compliance with External Requirements

Learning Outcomes

- Monitor governance effectiveness
 - Evaluate compliance and internal controls
-

Module 9: COBIT Performance Management

Topics Covered

- Introduction to Performance Management

- Capability Levels
- Maturity Measurement Concepts
- Process Performance Measurement
- Governance System Performance
- Metrics and KPIs

Learning Outcomes

- Measure governance process performance
 - Assess capability and maturity levels
-

Module 10: Design Factors

Topics Covered

- Enterprise Strategy
- Enterprise Goals
- Risk Profile
- Compliance Requirements
- Threat Landscape
- Role of IT
- Sourcing Model
- Technology Adoption Strategy
- Enterprise Size

Learning Outcomes

- Understand governance system customization
 - Apply design factors to enterprise scenarios
-

Module 11: Focus Areas

Topics Covered

- Information Security
- Digital Transformation
- Small and Medium Enterprises

- DevOps
- Cloud Computing
- Privacy Management
- Cybersecurity Governance
- Risk Governance

Learning Outcomes

- Tailor COBIT for specific business needs
 - Apply focus areas to real-world governance challenges
-

Module 12: Goals Cascade Mechanism

Topics Covered

- Stakeholder Drivers
- Stakeholder Needs
- Enterprise Goals
- Alignment Goals
- Governance and Management Objectives
- Goal Mapping Techniques

Learning Outcomes

- Align business goals with IT objectives
 - Perform goals cascade analysis
-

Module 13: COBIT Implementation Approach

Topics Covered

- Implementation Lifecycle
- Governance Improvement Program
- Change Management
- Continuous Improvement
- Governance Roadmap Development
- Common Implementation Challenges

Learning Outcomes

- Plan a COBIT implementation project
 - Manage governance transformation initiatives
-

Module 14: COBIT and Other Frameworks

Topics Covered

- COBIT and ITIL
- COBIT and ISO/IEC 27001
- COBIT and NIST Cybersecurity Framework
- COBIT and TOGAF
- COBIT and PMI
- COBIT and COSO

Learning Outcomes

- Integrate COBIT with other governance frameworks
 - Understand complementary framework relationships
-

Module 15: COBIT 2019 Foundation Exam Preparation

Topics Covered

- COBIT Terminology Review
- Governance Principles Review
- Domains and Objectives Revision
- Design Factors Review
- Performance Management Review
- Focus Areas Review
- Practice Questions
- Mock Tests
- Exam Strategies

Learning Outcomes

- Prepare for the COBIT 2019 Foundation Certification Exam

- Develop confidence in applying COBIT concepts
-

Recommended Course Duration

Module	Hours
Module 1–4	8 Hours
Module 5–8	12 Hours
Module 9–12	8 Hours
Module 13–14	4 Hours
Module 15	4 Hours
Total	36 Hours

Hands-On Activities

Activity 1

Map enterprise business goals to COBIT alignment goals.

Activity 2

Perform a governance gap assessment for a sample organization.

Activity 3

Design a governance system using COBIT design factors.

Activity 4

Assess process capability using COBIT Performance Management.

Activity 5

Develop a COBIT implementation roadmap for a fictional enterprise.

Capstone Project

Create a complete COBIT 2019 Governance Framework for an organization, including:

- Goals Cascade
- Governance Objectives
- Risk Assessment

- Performance Metrics
- Governance Dashboard
- Implementation Roadmap
- Compliance Mapping (ISO 27001/NIST/ITIL)

COBIT 2019 Design and Implementation

Course Overview

The COBIT 2019 Design & Implementation course focuses on designing a tailored governance system and implementing Enterprise Governance of Information and Technology (EGIT) using the COBIT 2019 framework. The curriculum aligns with the official **ISACA COBIT 2019 Design and Implementation** certification and prepares learners to assess, design, build, and sustain governance systems within organizations.

Module 1: Introduction to COBIT 2019 Design & Implementation

Topics Covered

- Overview of Enterprise Governance of Information and Technology (EGIT)
- COBIT 2019 Governance Framework Recap
- Purpose of Design & Implementation
- Business Drivers for Governance
- Governance Challenges in Modern Enterprises
- Relationship between Governance and Digital Transformation
- COBIT Design Guide and Implementation Guide Overview

Learning Outcomes

- Understand the purpose of COBIT Design & Implementation.
 - Explain how governance contributes to business value.
 - Identify governance challenges and opportunities.
-

Module 2: Understanding Governance System Design

Topics Covered

- Governance System Concepts
- Design Factors Overview
- Tailoring Governance Systems
- Governance Components
- Enterprise Context Assessment
- Stakeholder Requirements Analysis
- Governance Prioritization

Learning Outcomes

- Understand governance system design principles.
 - Identify factors affecting governance design.
 - Analyze organizational governance requirements.
-

Module 3: COBIT 2019 Design Factors

Topics Covered

1. Enterprise Strategy
2. Enterprise Goals
3. Risk Profile
4. IT-Related Issues
5. Threat Landscape
6. Compliance Requirements
7. Role of IT
8. Sourcing Model
9. IT Implementation Methods
10. Technology Adoption Strategy
11. Enterprise Size

Practical Exercise

- Assess design factors for a sample organization.

Learning Outcomes

- Evaluate enterprise characteristics.
 - Apply design factors to governance design.
-

Module 4: Governance System Tailoring

Topics Covered

- Designing Governance Priorities
- Determining Governance Objectives
- Customizing Governance Components
- Mapping Enterprise Goals
- Aligning IT and Business Objectives
- Governance System Architecture

Practical Exercise

- Create a customized governance blueprint.

Learning Outcomes

- Tailor governance systems to enterprise needs.
 - Develop governance architectures.
-

Module 5: Focus Areas and Governance Customization

Topics Covered

- Cybersecurity Governance
- Digital Transformation Governance
- Cloud Governance
- DevOps Governance
- Privacy Governance
- Risk Governance
- SME Governance Models

Practical Exercise

- Select and integrate relevant focus areas.

Learning Outcomes

- Apply focus areas to governance systems.
 - Customize governance for emerging technologies.
-

Module 6: COBIT Governance Design Workflow

Topics Covered

- Design Workflow Steps
- Stakeholder Analysis
- Governance Prioritization
- Design Toolkit Usage
- Governance Score Interpretation
- Target Capability Determination

Practical Exercise

- Build governance workflows using design tools.

Learning Outcomes

- Execute governance design methodologies.
 - Prioritize governance objectives effectively.
-

Module 7: Introduction to COBIT Implementation

Topics Covered

- COBIT Implementation Guide
- Governance Improvement Lifecycle
- Organizational Readiness Assessment
- Change Management Fundamentals
- Governance Transformation Programs
- Business Case Development

Learning Outcomes

- Understand governance implementation concepts.
- Develop implementation readiness assessments.

Module 8: COBIT Implementation Lifecycle – Phase 1 & 2

Phase 1: What Are the Drivers?

Topics Covered

- Pain Points Analysis
- Trigger Events
- Stakeholder Needs
- Governance Challenges

Phase 2: Where Are We Now?

Topics Covered

- Current State Assessment
- Capability Assessment
- Gap Analysis
- Risk Identification

Practical Exercise

- Conduct governance maturity assessment.

Learning Outcomes

- Identify governance drivers.
- Assess current governance maturity.

Module 9: COBIT Implementation Lifecycle – Phase 3 & 4

Phase 3: Where Do We Want To Be?

Topics Covered

- Target State Definition
- Governance Goals
- Desired Capabilities
- Strategic Alignment

Phase 4: What Needs To Be Done?

Topics Covered

- Solution Identification
- Governance Improvement Initiatives
- Roadmap Planning
- Resource Planning

Practical Exercise

- Develop governance target-state roadmap.

Learning Outcomes

- Define governance objectives.
 - Build governance implementation plans.
-

Module 10: COBIT Implementation Lifecycle – Phase 5

Phase 5: How Do We Get There?

Topics Covered

- Project Execution
- Governance Program Management
- Change Enablement
- Stakeholder Communication
- Training and Awareness

Practical Exercise

- Build implementation execution plans.

Learning Outcomes

- Execute governance transformation programs.
 - Manage governance change initiatives.
-

Module 11: COBIT Implementation Lifecycle – Phase 6

Phase 6: Did We Get There?

Topics Covered

- Benefits Realization
- KPI Monitoring

- Governance Metrics
- Performance Evaluation
- Post-Implementation Review

Practical Exercise

- Develop governance scorecards.

Learning Outcomes

- Measure implementation success.
 - Evaluate governance performance.
-

Module 12: COBIT Implementation Lifecycle – Phase 7

Phase 7: How Do We Keep the Momentum Going?

Topics Covered

- Continuous Improvement
- Governance Sustainability
- Lessons Learned
- Continuous Monitoring
- Governance Culture Development

Practical Exercise

- Design continuous improvement frameworks.

Learning Outcomes

- Sustain governance programs.
 - Establish governance improvement cycles.
-

Module 13: Organizational Change Management

Topics Covered

- Governance Transformation Challenges
- Change Resistance
- Stakeholder Engagement
- Communication Strategy

- Leadership Alignment
- Governance Culture

Learning Outcomes

- Manage organizational resistance.
 - Drive governance adoption.
-

Module 14: Measuring Governance Performance

Topics Covered

- COBIT Performance Management
- Capability Levels
- Maturity Models
- Metrics and KPIs
- Governance Dashboards
- Reporting Frameworks

Practical Exercise

- Create governance performance dashboards.

Learning Outcomes

- Assess governance effectiveness.
 - Develop governance measurement systems.
-

Module 15: Risk, Compliance and Assurance Integration

Topics Covered

- Governance and Risk Alignment
- Compliance Monitoring
- Internal Controls
- Audit Readiness
- Assurance Activities
- Third-Party Governance

Learning Outcomes

- Integrate governance with risk and compliance.
 - Prepare organizations for audits.
-

Module 16: COBIT Integration with Other Frameworks

Topics Covered

- COBIT and ITIL 4
- COBIT and ISO/IEC 27001
- COBIT and NIST Cybersecurity Framework
- COBIT and COSO
- COBIT and TOGAF
- COBIT and PMBOK

Learning Outcomes

- Align COBIT with enterprise frameworks.
 - Build integrated governance ecosystems.
-

Module 17: Governance Design Workshop

Hands-On Activities

- Analyze a Case Study Organization
- Identify Design Factors
- Create Governance Objectives Profile
- Prioritize Governance Components
- Build Governance Architecture

Deliverables

- Governance Design Document
 - Governance Objectives Matrix
 - Governance Capability Assessment
-

Module 18: Governance Implementation Workshop

Hands-On Activities

- Conduct Current State Assessment
- Build Target State Model
- Create Governance Roadmap
- Develop Change Management Plan
- Define KPIs and Success Metrics

Deliverables

- Implementation Roadmap
 - Governance Dashboard
 - Benefits Realization Plan
-

Module 19: Capstone Project

Enterprise Governance Transformation Project

Project Tasks

- Analyze Enterprise Context
- Assess Design Factors
- Design Governance System
- Select Governance Objectives
- Develop Implementation Roadmap
- Create Governance Metrics
- Establish Continuous Improvement Strategy

Final Deliverables

- COBIT Governance Design Blueprint
 - Governance Objectives Profile
 - Risk and Compliance Mapping
 - Governance Dashboard
 - Implementation Roadmap
 - Executive Presentation
-

Module 20: COBIT Design & Implementation Certification Preparation

Topics Covered

- Exam Domain Review
- Design Factors Revision
- Implementation Lifecycle Revision
- Governance Scenarios
- Case Study Analysis
- Practice Questions
- Mock Exams
- Exam Strategies

Learning Outcomes

- Prepare for COBIT Design & Implementation certification.
 - Apply governance concepts to real-world scenarios.
-
- **CGRC- Domain 1: Security and Privacy Governance, Risk Management, and Compliance Program (16%)**
 - 1.1 – Demonstrate knowledge in security and privacy governance, risk management, and compliance program
 - Principles of governance, risk management, and compliance
 - Risk management and compliance frameworks using national and international standards and guidelines for security and privacy requirements (e.g., National Institute of Standards and Technology (NIST), cybersecurity framework, Control Objectives for Information and Related Technology (COBIT), International Organization for Standardization/International Electrotechnical Commission (ISO/IEC))
 - System Development Life Cycle (SDLC) (e.g., requirements gathering, design, development, testing, and operations/maintenance/disposal)
 - Information lifecycle for each data type processed, stored, or transmitted (e.g., retaining, disposal/destruction, data flow, marking)

- Confidentiality, integrity, availability, non-repudiation, and privacy concepts
- System assets and boundary descriptions
- Security and privacy controls and requirements
- Roles and responsibilities for compliance activities and associated frameworks
- 1.2 – Demonstrate knowledge in security and privacy governance, risk management, and compliance program processes
 - Establishment of compliance program for the applicable framework
- 1.3 – Understand regulatory and legal requirements
- 1.3 – Demonstrate knowledge of compliance frameworks, regulations, privacy, and security requirements
 - Familiarity with compliance frameworks (e.g., International Organization for Standardization/International Electrotechnical Commission (ISO/IEC), Federal Risk and Authorization Management Program (FedRAMP), Payment Card Industry Data Security Standard (PCI-DSS), Cybersecurity Maturity Model Certification)
 - Familiarity with other national and international laws and requirements for security and privacy (e.g., Federal Information Security Modernization Act (FISMA), Health Insurance Portability and Accountability Act (HIPAA), executive orders, General Data Protection Regulation (GDPR))
- **Domain 2: Scope of the System (10%)**
 - 2.1 – Describe the system
 - System name and scope documented
 - System purpose and functionality
 - 2.2 – Determine security compliance required
 - Information types processed, stored, or transmitted
 - Security objectives outlined for each information type based on national and international security and privacy compliance requirements (e.g., Federal Information Processing Standards (FIPS), International Organization for Standardization/International Electrotechnical Commission (ISO/IEC), data protection impact assessment)

- Risk impact level determined for system based on the selected framework
- **Domain 3: Selection and Approval of Framework, Security, and Privacy Controls (14%)**
 - 3.1 – Identify and document baseline and inherited controls
 - 3.2 – Select and tailor controls
 - Determination of applicable baseline and/or inherited controls
 - Determination of appropriate control enhancements (e.g., security practices, overlays, mitigating controls)
 - Specific data handling/marking requirements identified
 - Control selection documentation
 - Continued compliance strategy (e.g., continuous monitoring, vulnerability management)
 - Control allocation and stakeholder agreement
- **Domain 4: Implementation of Security and Privacy Controls (17%)**
 - 4.1 – Develop implementation strategy (e.g., resourcing, funding, timeline, effectiveness)
 - Control implementation aligned with organizational expectations, national or international requirements, and compliance for security and privacy controls
 - Identification of control types (e.g., management, technical, common, operational control)
 - Frequency established for compliance documentation reviews and training
 - 4.2 – Implement selected controls
 - Control implementation consistent with compliance requirements
 - Compensating or alternate security controls implemented
 - 4.3 – Document control implementation
 - Residual security risk or planned implementations documented (e.g., Plan of Action and Milestones (POA&M), risk register)
 - Implemented controls documented consistent with the organization's purpose, scope, and risk profile (e.g., policies, procedures, plans)

- **Domain 5: Assessment/Audit of Security and Privacy Controls (16%)**
 - 5.1 – Prepare for assessment/audit
 - Stakeholder roles and responsibilities established
 - Objectives, scope, resources, schedule, deliverables, and logistics outlined
 - Assets, methods, and level of effort scoped
 - Evidence for demonstration of compliance audited (e.g., previous assessments/audits, system documentation, policies)
 - Assessment/audit plan finalized
 - 5.2 – Conduct assessment/audit
 - Compliance capabilities verified using appropriate assessment methods: interview, examine, test (e.g., penetration, control, vulnerability scanning)
 - Evidence verified and validated
 - 5.3 – Prepare the initial assessment/audit report
 - Risks identified during the assessment/audit provided
 - Risk mitigation summaries outlined
 - Preliminary findings recorded
 - 5.4 – Review initial assessment/audit report and plan risk response actions
 - Risk response assigned (e.g., avoid, accept, share, mitigate, transfer) based on identified vulnerabilities or deficiencies
 - Risk response collaborated with stakeholders
 - Non-compliant findings with newly applied corrective actions reassessed and validated
 - 5.5 – Develop final assessment/audit report
 - Final compliance documented (e.g., compliant, non-compliant, not applicable)
 - Recommendations documented when appropriate
 - Assessment report finalized
 - 5.6 – Develop risk response plan
 - Residual risks and deficiencies identified

- Risk prioritized
- Required resources identified (e.g., financial, personnel, and technical) to determine time required to mitigate risk
- **Domain 6: System Compliance (14%)**
 - 6.1 – Review and submit security/privacy documents
 - Security and privacy documentation required to support a compliance decision by the appropriate party (e.g., authorizing official, third-party assessment organizations, agency) compiled, reviewed, and submitted
 - 6.2 – Determine system risk posture
 - System risk acceptance criteria
 - Residual risk determination
 - Stakeholder concurrence for risk treatment options
 - Residual risks defined in formal documentation
 - 6.3 – Document system compliance
 - Formal notification of compliance decision
 - Formal notification shared with stakeholders
- **Domain 7: Compliance Maintenance (13%)**
 - 7.1 – Perform system change management
 - Changes weigh the impact to organizational risk, operations, and/or compliance requirements (e.g., revisions to baselines)
 - Proposed changes documented and approved by authorized personnel (e.g., Change Control Board (CCB), technical review board)
 - Deploy to the environment (e.g., test, development, production) with rollback plan
 - Changes to the system tracked and compliance enforced
 - 7.2 – Perform ongoing compliance activities based on requirements
 - Frequency established for ongoing compliance activities review with stakeholders
 - System and assets monitored (e.g., physical and logical assets, personnel, change control)

- Incident response and contingency activities performed
- Security updates performed and risks remediated/tracked
- Evidence collected, testing performed, documentation updated (e.g., service level agreements, third party contracts, policies, procedures), and submission/communication to stakeholders when applicable
- Awareness and training performed, documented, and retained (e.g., contingency, incident response, annual security, and privacy)
- Revising monitoring strategies based on updates to legal, regulatory, supplier, security and privacy requirements
- 7.3 – Engage in audit activities based on compliance requirements
 - Required testing and vulnerability scanning performed
 - Personnel interviews conducted
 - Documentation reviewed and updated
- 7.4 – Decommission system when applicable
 - Requirements for system decommissioning reviewed with stakeholders
 - System removed from operations and decommissioned
 - Documentation of the decommissioned system retained and shared with stakeholders

ISO/IEC 20000 Implementer

Module 1: Introduction to IT Service Management (ITSM)

Topics Covered

- Fundamentals of IT Service Management
- Evolution of ITSM
- Business Value of IT Services
- IT Service Management Concepts
- Service Lifecycle Overview

- Service Provider Roles and Responsibilities
- Relationship between ITSM and Business Strategy

Learning Outcomes

- Understand ITSM principles and terminology.
 - Explain the value of service management in organizations.
 - Identify the components of an effective SMS.
-

Module 2: Introduction to ISO/IEC 20000

Topics Covered

- Overview of ISO/IEC 20000 Family
- Purpose and Benefits of Certification
- ISO/IEC 20000-1 Requirements
- ISO/IEC 20000-2 Guidance
- Structure of the Standard
- Annex SL Framework
- Relationship with Other Standards

Learning Outcomes

- Understand ISO/IEC 20000 requirements.
 - Interpret clauses of the standard.
 - Explain certification benefits.
-

Module 3: Service Management System (SMS) Fundamentals

Topics Covered

- Service Management System Architecture
- Scope Definition
- SMS Policies and Objectives
- Process-Based Approach
- Service Value Creation
- Stakeholder Expectations

- Service Governance

Learning Outcomes

- Design a Service Management System.
 - Establish SMS scope and objectives.
 - Align SMS with business needs.
-

Module 4: Organizational Context and Leadership

Topics Covered

Clause 4 – Context of the Organization

- Internal and External Issues
- Interested Parties
- Scope Determination
- SMS Processes

Clause 5 – Leadership

- Leadership Commitment
- Service Management Policy
- Roles and Responsibilities
- Governance Structures

Learning Outcomes

- Analyze organizational context.
 - Define leadership responsibilities.
 - Establish governance mechanisms.
-

Module 5: Planning the Service Management System

Topics Covered

Clause 6 – Planning

- Risk-Based Thinking
- Opportunities Management
- Service Management Objectives

- Change Planning
- Strategic Service Planning
- Resource Planning

Practical Exercise

- Create an SMS implementation plan.

Learning Outcomes

- Plan SMS implementation.
 - Identify risks and opportunities.
 - Develop measurable objectives.
-

Module 6: Support Processes for SMS

Topics Covered

Clause 7 – Support

- Resource Management
- Competence and Training
- Awareness Programs
- Communication Management
- Documented Information
- Knowledge Management
- Service Documentation Control

Practical Exercise

- Develop document control procedures.

Learning Outcomes

- Establish support processes.
 - Manage documentation and records.
 - Develop staff competency plans.
-

Module 7: Service Portfolio and Service Level Management

Topics Covered

- Service Portfolio Management
- Service Catalog Management
- Service Level Agreements (SLAs)
- Operational Level Agreements (OLAs)
- Service Reporting
- Customer Relationship Management
- Supplier Relationship Management

Practical Exercise

- Create SLA and Service Catalog templates.

Learning Outcomes

- Manage service portfolios effectively.
 - Define and monitor service levels.
-

Module 8: Service Design and Transition

Topics Covered

- Service Design Principles
- Service Transition Management
- Release and Deployment Management
- Change Enablement
- Configuration Management
- Asset Management
- Capacity Planning

Practical Exercise

- Design a new IT service implementation plan.

Learning Outcomes

- Implement service design processes.
 - Manage service transitions effectively.
-

Module 9: Incident and Service Request Management

Topics Covered

- Incident Management
- Service Request Fulfillment
- Major Incident Handling
- Escalation Procedures
- Service Desk Operations
- Incident Prioritization
- Incident Reporting

Practical Exercise

- Simulate incident response workflows.

Learning Outcomes

- Implement incident management processes.
 - Improve service responsiveness.
-

Module 10: Problem Management

Topics Covered

- Problem Identification
- Root Cause Analysis
- Known Error Database
- Preventive Actions
- Trend Analysis
- Problem Resolution Workflows

Practical Exercise

- Conduct Root Cause Analysis (RCA).

Learning Outcomes

- Reduce recurring incidents.
 - Implement structured problem management.
-

Module 11: Availability, Capacity and Continuity Management

Topics Covered

- Availability Management
- Capacity Management
- IT Service Continuity Management
- Disaster Recovery Planning
- Business Impact Analysis
- Resilience Strategies

Practical Exercise

- Develop continuity and recovery plans.

Learning Outcomes

- Improve service resilience.
 - Ensure service continuity.
-

Module 12: Information Security and Risk Management

Topics Covered

- Information Security in ITSM
- Service Risks
- Security Controls
- Risk Assessment Methodologies
- Supplier Risks
- Compliance Requirements
- Security Incident Handling

Practical Exercise

- Perform service risk assessments.

Learning Outcomes

- Integrate security into ITSM.
 - Manage service-related risks.
-

Module 13: Performance Evaluation

Topics Covered

Clause 9 – Performance Evaluation

- Service Metrics and KPIs
- Monitoring and Measurement
- Internal Audits
- Management Reviews
- Customer Satisfaction Analysis
- Performance Reporting

Practical Exercise

- Create KPI dashboards.

Learning Outcomes

- Measure SMS performance.
 - Conduct management reviews.
-

Module 14: Continual Improvement

Topics Covered

Clause 10 – Improvement

- Continual Improvement Process
- Corrective Actions
- Nonconformity Management
- PDCA Cycle
- Service Optimization
- Lessons Learned Programs

Practical Exercise

- Develop continual improvement plans.

Learning Outcomes

- Implement improvement initiatives.
 - Enhance service maturity.
-

Module 15: ISO/IEC 20000 Implementation Methodology

Topics Covered

- PECB Implementation Framework
- SMS Implementation Lifecycle
- Gap Assessment
- Project Planning
- Process Development
- Pilot Deployment
- Full-Scale Implementation

Practical Exercise

- Build an SMS implementation roadmap.

Learning Outcomes

- Execute structured SMS implementations.
 - Manage implementation projects effectively.
-

Module 16: ITSM Governance and Integration

Topics Covered

- ITSM Governance Framework
- Integration with IT Governance
- Service Governance Models
- Organizational Roles
- Service Ownership
- Governance Committees

Learning Outcomes

- Establish ITSM governance structures.
 - Align IT services with organizational objectives.
-

Module 17: Integration with Other Standards

Topics Covered

- ISO/IEC 20000 and ISO 9001
- ISO/IEC 20000 and ISO/IEC 27001
- ISO/IEC 20000 and ISO 22301
- ISO/IEC 20000 and COBIT 2019
- ISO/IEC 20000 and ITIL 4
- Governance and Compliance Alignment

Learning Outcomes

- Integrate SMS with other management systems.
 - Build unified governance models.
-

Module 18: Internal Audit and Certification Preparation

Topics Covered

- Audit Principles
- Audit Planning
- Audit Checklists
- Evidence Collection
- Nonconformity Reporting
- Certification Process
- Readiness Assessment

Practical Exercise

- Conduct a mock certification audit.

Learning Outcomes

- Prepare organizations for certification audits.
 - Evaluate SMS compliance.
-

Module 19: Service Management System Implementation Workshop

Hands-On Activities

- Define SMS Scope
- Conduct Gap Analysis

- Develop SMS Policy
- Create Service Catalog
- Design Incident Management Process
- Develop KPI Framework
- Build Audit Plan

Deliverables

- SMS Implementation Plan
 - Service Catalog
 - KPI Dashboard
 - Audit Checklist
-

Module 20: Capstone Project

End-to-End ISO/IEC 20000 SMS Implementation

Project Tasks

- Organizational Context Analysis
- SMS Scope Definition
- Risk Assessment
- Service Catalog Design
- SLA Development
- Incident & Problem Management Design
- Performance Measurement Framework
- Internal Audit Planning
- Continual Improvement Roadmap

Final Deliverables

- Service Management System Manual
- SMS Policy and Objectives
- Service Catalog
- SLA Framework
- Risk Register

- KPI Dashboard
 - Internal Audit Program
 - Certification Readiness Report
-

Module 21: PECB ISO/IEC 20000 Lead Implementer Exam Preparation

Topics Covered

- ISO/IEC 20000 Clause Review
- SMS Requirements Review
- Implementation Lifecycle Review
- Service Management Processes
- Governance and Compliance
- Scenario-Based Questions
- Mock Examinations
- Exam Strategies

Learning Outcomes

- Prepare for the Lead Implementer certification exam.
- Apply ISO/IEC 20000 requirements in real-world scenarios.
-

CGEIT- **Domain 1:** Framework for the Governance of Enterprise IT.

- **Domain 2:** Strategic Management.
- **Domain 3:** Benefits Realization.
- **Domain 4:** Risk Optimization.
- **Domain 5:** Resource Optimization.

Introduction to the Information Security Management System (ISMS) and ISO/IEC 27001

Section 1: Training course objectives and structure

- General information
- Learning objectives
- Educational approach
- Examination and certification

Section 2: Standards and regulatory frameworks

- What is ISO?
- The ISO/IEC 27000 family of standards
- Advantages of ISO/IEC 27001

Section 3: Certification process

- Certification process
- Certification scheme
- Accreditation bodies
- Certification bodies

Section 4: Fundamental concepts and principles of information security

- Information and asset
- Information security
- Confidentiality, integrity, and availability
- Vulnerability, threat, and impact
- Information security risk
- Security controls and control objectives
- Classification of security controls

Section 5: Information Security Management System (ISMS)

- Definition of a management system
- Definition of ISMS
- Process approach
- ISMS implementation
- Overview – Clauses 4 to 10
- Overview – Annex A
- Statement of Applicability

Audit principles, preparation, and initiation of an audit

Section 6: Fundamental audit concepts and principles

- Audit standards
- What is an audit?
- Types of audits
- Involved parties
- Audit objectives and criteria
- Combined audit
- Principles of auditing
- Competence and evaluation of auditors

Section 7: The impact of trends and technology in auditing

- Big data
- The three V's of big data
- The use of big data in audits
- Artificial intelligence
- Machine learning
- Cloud computing
- Auditing outsourced operations

Section 8: Evidence-based auditing

- Audit evidence
- Types of audit evidence
- Quality and reliability of audit evidence

Section 9: Risk-based auditing

- Audit approach based on risk
- Materiality and audit planning
- Reasonable assurance

Section 10: Initiation of the audit process

- The audit offer
- The audit team leader
- The audit team
- Audit feasibility
- Audit acceptance
- Establishing contact with the auditee
- The audit schedule

Section 11: Stage 1 audit

- Objectives of the stage 1 audit
- Pre on-site activities
- Preparing for on-site activities
- Conducting on-site activities
- Documenting the outputs of stage 1 audit

On-site audit activities

Section 12: Preparing for stage 2 audit

- Setting the audit objectives
- Planning the audit

- Assigning work to the audit team
- Preparing audit test plans
- Preparing documented information for the audit

Section 13: Stage 2 audit

- Conducting the opening meeting
- Collecting information
- Conducting audit tests
- Determining audit findings and nonconformity reports
- Performing a quality review

Section 14: Communication during the audit

- Behavior during on-site visits
- Communication during the audit
- Audit team meetings
- Guides and observers
- Conflict management
- Cultural aspects
- Communication with the top management

Section 15: Audit procedures

- Overview of the audit process
- Evidence collection and analysis procedures
- Interview
- Documented information review
- Observation
- Analysis
- Sampling
- Technical verification

Section 16: Creating audit test plans

- Audit test plans
- Examples of audit test plans
- Guidance for auditing an ISMS
- Corroboration
- Evaluation
- Auditing virtual activities and locations

Closing of the audit

Section 17: Drafting audit findings and nonconformity reports

- Audit findings
- Types of possible audit findings
- Documenting the audit findings
- Drafting a nonconformity report
- The principle of the benefit of the doubt

Section 18: Audit documentation and quality review

- Work documents
- Quality review

Section 19: Closing of the audit

- Determining audit conclusions
- Discussing audit conclusions
- Closing meeting
- Preparing audit report
- Distributing the audit report
- Making the certification decision
- Closing the audit

Section 20: Evaluation of action plans by the auditor

- Submission of action plans by the auditee
- Content of action plans
- Evaluation of action plans

Section 21: Beyond the initial audit

- Audit follow-up activities
- Surveillance activities
- Recertification audit
- Use of trademarks

Section 22: Managing an internal audit program

- Managing an audit program
- Role of the internal audit function
- Main internal audit services and activities
- Audit program resources
- Audit program records
- Follow up on nonconformities
- Monitoring, evaluating, reviewing, and improving an audit program

The above-mentioned content is delivered in 32 hours. In addition to this, we have added 8 hours session.

8hrs dedicated session

ISO 27001 Practical Approach

- ISO 27001 (new 93 controls) Controls to Evidence Mapping
- Practical approach on how to collect evidence while auditing with three scenarios/ case studies paragraphs

ISO 27001 Exam Prep

- Revision of course and open mic session for doubts
- Exam Prep – mock exam
- Discussion on exam questions and answers
- Discussion on different exams

CCNA- Introduction to Networking

Chapter 1: Introduction to TCP/IP Networking

- TCP/IP Networking Model
- History Leading to TCP/IP
- Overview of the TCP/IP Networking Model
- TCP/IP Application Layer
- HTTP Overview
- HTTP Protocol Mechanisms
- TCP/IP Transport Layer
- TCP Error Recovery Basics
- Same-Layer and Adjacent-Layer Interactions
- TCP/IP Network Layer
- Internet Protocol and the Postal Service
- Internet Protocol Addressing Basics
- IP Routing Basics
- TCP/IP Data-Link and Physical Layers
- Data Encapsulation Terminology
- Names of TCP/IP Messages
- OSI Networking Model and Terminology
- Comparing OSI and TCP/IP Layer Names and Numbers
- OSI Data Encapsulation Terminology

Chapter 2: Fundamentals of Ethernet LANs

- An Overview of LANs
- SOHO LANs
- Enterprise LANs
- The Variety of Ethernet Physical Layer Standards
- Consistent Behavior over All Links Using the Ethernet Data-Link Layer
- Building Physical Ethernet LANs with UTP
- Transmitting Data Using Twisted Pairs
- Breaking Down a UTP Ethernet Link
- UTP Cabling Pinouts for 10BASE-T and 100BASE-T
- Straight-Through Cable Pinout
- Choosing the Right Cable Pinouts
- UTP Cabling Pinouts for 1000BASE-T
- Building Physical Ethernet LANs with Fiber
- Fiber Cabling Transmission Concepts
- Using Fiber with Ethernet
- Sending Data in Ethernet Networks
- Ethernet Data-Link Protocols
- Ethernet Addressing
- Identifying Network Layer Protocols with the Ethernet Type Field
- Error Detection with FCS
- Sending Ethernet Frames with Switches and Hubs
- Sending in Modern Ethernet LANs Using Full Duplex
- Using Half Duplex with LAN Hubs

Chapter 3: Fundamentals of WANs and IP Routing

- Wide-Area Networks
- Leased-Line WANs
- Physical Details of Leased Lines
- HDLC Data-Link Details of Leased Lines
- How Routers Use a WAN Data Link

- Ethernet as a WAN Technology
- Ethernet WANs That Create a Layer 2 Service
- How Routers Route IP Packets Using Ethernet Emulation
- IP Routing
- Network Layer Routing (Forwarding) Logic
- Host Forwarding Logic: Send the Packet to the Default Router
- R1 and R2's Logic: Routing Data Across the Network
- R3's Logic: Delivering Data to the End Destination
- How Network Layer Routing Uses LANs and WAN
- How IP Addressing Helps IP Routing
- Rules for Groups of IP Addresses (Networks and Subnets)
- The IP Header
- How IP Routing Protocols Help IP Routing
- Other Network Layer Features
- Using Names and the Domain Name System
- The Address Resolution Protocol
- ICMP Echo and the ping Command

Implementing Ethernet LANs

Chapter 4: Using the Command-Line Interface

- Accessing the Cisco Catalyst Switch CLI
- Cisco Catalyst Switches
- Accessing the Cisco IOS CLI
- Cabling the Console Connection
- Accessing the CLI with Telnet and SSH
- User and Enable (Privileged) Modes
- Password Security for CLI Access from the Console
- CLI Help Features
- The debug and show Commands
- Configuring Cisco IOS Software

- Configuration Submodels and Contexts
- Storing Switch Configuration Files
- Copying and Erasing Configuration Files

Chapter 5: Analysing Ethernet LAN Switching

- LAN Switching Concepts
- Overview of Switching Logic
- Forwarding Known Unicast Frames
- Learning MAC Addresses
- Flooding Unknown Unicast and Broadcast Frames
- Avoiding Loops Using Spanning Tree Protocol
- LAN Switching Summary
- Verifying and Analyzing Ethernet Switching
- Demonstrating MAC Learning
- Switch Interfaces
- Finding Entries in the MAC Address Table
- Managing the MAC Address Table (Aging, Clearing)
- MAC Address Tables with Multiple Switches

Chapter 6: Configuring Basic Switch Management

- Securing the Switch CLI
- Securing User Mode and Privileged Mode with Simple Passwords
- Securing User Mode Access with Local Usernames and Passwords
- Securing User Mode Access with External Authentication Servers
- Securing Remote Access with Secure Shell
- Enabling IPv4 for Remote Access
- Host and Switch IP Settings
- Configuring IPv4 on a Switch
- Configuring a Switch to Learn Its IP Address with DHCP
- Verifying IPv4 on a Switch
- Miscellaneous Settings Useful in the Lab

- History Buffer Commands
- The logging synchronous, exec-timeout, and no ip domain-lookup Commands

Chapter 7: Configuring and Verifying Switch Interfaces

- Configuring Switch Interfaces
- Configuring Speed, Duplex, and Description
- Configuring Multiple Interfaces with the interface range Command
- Administratively Controlling Interface State with shutdown
- Removing Configuration with the no Command
- Auto-negotiation
- Auto-negotiation Under Working Conditions
- Auto-negotiation Results When Only One Node Uses Autonegotiation
- Auto-negotiation and LAN Hubs
- Analyzing Switch Interface Status and Statistics
- Interface Status Codes and Reasons for Nonworking States
- Interface Speed and Duplex Issues
- Common Layer 1 Problem on Working Interfaces

Implementing VLANs and STP

Chapter 8: Implementing Ethernet Virtual LANs

- Virtual LAN Concepts
- Creating Multiswitch VLANs Using Trunking
- VLAN Tagging Concepts
- The 802.1Q and ISL VLAN Trunking Protocols
- Forwarding Data Between VLANs
- The Need for Routing Between VLANs
- Routing Packets Between VLANs with a Router
- VLAN and VLAN Trunking Configuration and Verification
- Creating VLANs and Assigning Access VLANs to an Interface
- VLAN Configuration Example 1: Full VLAN Configuration
- VLAN Configuration Example 2: Shorter VLAN Configuration

- VLAN Trunking Protocol
- VLAN Trunking Configuration
- Implementing Interfaces Connected to Phones
- Data and Voice VLAN Concepts
- Data and Voice VLAN Configuration and Verification
- Summary: IP Telephony Ports on Switches
- Troubleshooting VLANs and VLAN Trunks
- Access VLANs Undefined or Disabled
- Mismatched Trunking Operational State
- The Supported VLAN List on Trunks
- Mismatched Native VLAN on a Trunk

Chapter 9: Spanning Tree Protocol Concepts

- STP and RSTP Basics
- The Need for Spanning Tree
- What Spanning Tree Does
- How Spanning Tree Works
- The STP Bridge ID and Hello BPDU
- Electing the Root Switch
- Choosing Each Switch's Root Port
- Choosing the Designated Port on Each LAN Segment
- Configuring to Influence the STP Topology
- Details Specific to STP (and Not RSTP)
- STP Activity When the Network Remains Stable
- STP Timers That Manage STP Convergence
- Changing Interface States with STP
- Rapid STP Concepts
- Comparing STP and RSTP
- RSTP and the Alternate (Root) Port Role
- RSTP States and Processes

- RSTP and the Backup (Designated) Port Role
- RSTP Port Types
- Optional STP Features
- EtherChannel
- PortFast
- BPDU Guard

Chapter 10: RSTP and EtherChannel Configuration

- Understanding RSTP Through Configuration
- The Need for Multiple Spanning Trees
- STP Modes and Standards
- The Bridge ID and System ID Extension
- How Switches Use the Priority and System ID Extension
- RSTP Methods to Support Multiple Spanning Trees
- Other RSTP Configuration Options
- Configuring Layer 2 EtherChannel
- Configuring a Manual Layer 2 EtherChannel
- Configuring Dynamic EtherChannels
- Physical Interface Configuration and EtherChannels
- EtherChannel Load Distribution
- Configuration Options for EtherChannel Load Distribution
- The Effects of the EtherChannel Load Distribution Algorithm

IPv4 Addressing

Chapter 11: Perspectives on IPv4 Subnetting

- Introduction to Subnetting
- Subnetting Defined Through a Simple Example
- Operational View Versus Design View of Subnetting
- Analyze Subnetting and Addressing Needs
- Rules About Which Hosts Are in Which Subnet
- Determining the Number of Subnets

- Determining the Number of Hosts per Subnet
- One Size Subnet Fits All—Or Not
- Defining the Size of a Subnet
- One Size Subnet Fits All
- Multiple Subnet Sizes (Variable-Length Subnet Masks)
- One Mask for All Subnets, or More Than One
- Make Design Choices
- Choose a Classful Network
- Public IP Networks
- Growth Exhausts the Public IP Address Space
- Private IP Networks
- Choosing an IP Network During the Design Phase
- Choose the Mask
- Classful IP Networks Before Subnetting
- Borrowing Host Bits to Create Subnet Bits
- Choosing Enough Subnet and Host Bits
- Example Design: 172.16.0.0, 200 Subnets, 200 Hosts
- Masks and Mask Formats
- Build a List of All Subnets
- Plan the Implementation
- Assigning Subnets to Different Locations
- Choose Static and Dynamic Ranges per Subnet

Chapter 12: Analysing Classful IPv4 Networks

- Classful Network Concepts
- IPv4 Network Classes and Related Facts
- The Number and Size of Class A, B, and C Networks
- Address Formats
- Default Masks
- Number of Hosts per Network

- Deriving the Network ID and Related Numbers
- Unusual Network IDs and Network Broadcast Addresses
- Practice with Classful Networks
- Practice Deriving Key Facts Based on an IP Address
- Practice Remembering the Details of Address Classes

Chapter 13: Analyzing Subnet Masks

- Subnet Mask Conversion
- Three Mask Formats
- Converting Between Binary and Prefix Masks
- Converting Between Binary and DDN Masks
- Converting Between Prefix and DDN Masks
- Practice Converting Subnet Masks
- Identifying Subnet Design Choices Using Masks
- Masks Divide the Subnet's Addresses into Two Parts
- Masks and Class Divide Addresses into Three Parts
- Classless and Classful Addressing
- Calculations Based on the IPv4 Address Format
- Practice Analyzing Subnet Masks

Chapter 14: Analyzing Existing Subnets

- Defining a Subnet
- An Example with Network 172.16.0.0 and Four Subnets
- Subnet ID Concepts
- Subnet Broadcast Address
- Range of Usable Addresses
- Analyzing Existing Subnets: Binary
- Finding the Subnet ID: Binary
- Finding the Subnet Broadcast Address: Binary
- Binary Practice Problems
- Shortcut for the Binary Process

- Brief Note About Boolean Math
- Finding the Range of Addresses
- Analyzing Existing Subnets: Decimal
- Analysis with Easy Masks
- Predictability in the Interesting Octet
- Finding the Subnet ID: Difficult Masks
- Resident Subnet Example 1
- Resident Subnet Example 2
- Resident Subnet Practice Problems
- Finding the Subnet Broadcast Address: Difficult Masks
- Subnet Broadcast Example 1
- Subnet Broadcast Example 2
- Subnet Broadcast Address Practice Problems
- Practice Analyzing Existing Subnets
- A Choice: Memorize or Calculate

IPv4 Routing

Chapter 15: Operating Cisco Routers

- Installing Cisco Routers
- Installing Enterprise Routers
- Cisco Integrated Services Routers
- Physical Installation
- Installing SOHO Routers
- Enabling IPv4 Support on Cisco Router Interfaces
- Accessing the Router CLI
- Router Interfaces
- Interface Status Codes
- Router Interface IP Addresses
- Bandwidth and Clock Rate on Serial Interfaces
- Router Auxiliary Port

Chapter 16: Configuring IPv4 Addresses and Static Routes

- IP Routing
- IPv4 Routing Process Reference
- An Example of IP Routing
- Host Forwards the IP Packet to the Default Router (Gateway)
- Routing Step 1: Decide Whether to Process the Incoming Frame
- Routing Step 2: De-encapsulation of the IP Packet
- Routing Step 3: Choosing Where to Forward the Packet
- Routing Step 4: Encapsulating the Packet in a New Frame
- Routing Step 5: Transmitting the Frame
- Configuring IP Addresses and Connected Routes
- Connected Routes and the ip address Command
- The ARP Table on a Cisco Router
- Configuring Static Routes
- Static Network Routes
- Static Host Routes
- Floating Static Routes
- Static Default Routes
- Troubleshooting Static Routes
- Troubleshooting Incorrect Static Routes That Appear in the IP Routing Table
- The Static Route Does Not Appear in the IP Routing Table
- The Correct Static Route Appears but Works Poorly
- IP Forwarding with the Longest Prefix Match
- Using show ip route to Find the Best Route
- Using show ip route address to Find the Best Route
- Interpreting the IP Routing Table

Chapter 17: IP Routing in the LAN

- VLAN Routing with Router 802.1Q Trunks
- Configuring ROAS

- Verifying ROAS
- Troubleshooting ROAS
- VLAN Routing with Layer 3 Switch SVIs
- Configuring Routing Using Switch SVIs
- Verifying Routing with SVIs
- Troubleshooting Routing with SVIs
- VLAN Routing with Layer 3 Switch Routed Ports
- Implementing Routed Interfaces on Switches
- Implementing Layer 3 EtherChannels
- Troubleshooting Layer 3 EtherChannels

Chapter 18: Troubleshooting IPv4 Routing

- Problem Isolation Using the Ping Command
- Ping Command Basics
- Strategies and Results When Testing with the Ping Command
- Testing Longer Routes from Near the Source of the Problem
- Using Extended Ping to Test the Reverse Route
- Testing LAN Neighbors with Standard Ping
- Testing LAN Neighbors with Extended Ping
- Testing WAN Neighbors with Standard Ping
- Using Ping with Names and with IP Addresses
- Problem Isolation Using the Traceroute Command
- Traceroute Basics
- How the Traceroute Command Works
- Standard and Extended Traceroute
- Telnet and SSH
- Common Reasons to Use the IOS Telnet and SSH Client
- IOS Telnet and SSH Examples

OSPF

Chapter 19: Understanding OSPF Concepts

- Comparing Dynamic Routing Protocol Features
- Routing Protocol Functions
- Interior and Exterior Routing Protocols
- Comparing IGPs
- IGP Routing Protocol Algorithms
- Metrics
- Other IGP Comparisons
- Administrative Distance
- OSPF Concepts and Operation
- OSPF Overview
- Topology Information and LSAs
- Applying Dijkstra SPF Math to Find the Best Routes
- Becoming OSPF Neighbors
- The Basics of OSPF Neighbors
- Meeting Neighbors and Learning Their Router ID
- Exchanging the LSDB Between Neighbors
- Fully Exchanging LSAs with Neighbors
- Maintaining Neighbors and the LSDB
- Using Designated Routers on Ethernet Links
- Calculating the Best Routes with SPF
- OSPF Areas and LSAs
- OSPF Areas
- How Areas Reduce SPF Calculation Time
- (OSPFv2) Link-State Advertisements
- Router LSAs Build Most of the Intra-Area Topology
- Network LSAs Complete the Intra-Area Topology

Chapter 20: Implementing OSPF

- Implementing Single-Area OSPFv2
- OSPF Single-Area Configuration

- Wildcard Matching with the network Command
- Verifying OSPF Operation
- Verifying OSPF Configuration
- Configuring the OSPF Router ID
- Implementing Multiarea OSPF
- Using OSPFv2 Interface Subcommands
- OSPF Interface Configuration Example
- Verifying OSPF Interface Configuration
- Additional OSPFv2 Features
- OSPF Passive Interfaces
- OSPF Default Routes
- OSPF Metrics (Cost)
- Setting the Cost Directly
- Setting the Cost Based on Interface and Reference Bandwidth
- OSPF Load Balancing

Chapter 21: OSPF Network Types and Neighbors

- OSPF Network Types
- The OSPF Broadcast Network Type
- Verifying Operations with Network Type Broadcast
- Configuring to Influence the DR/BDR Election
- The OSPF Point-to-Point Network Type
- OSPF Neighbor Relationships
- OSPF Neighbor Requirements
- Issues That Prevent Neighbor Adjacencies
- Finding Area Mismatches
- Finding Duplicate OSPF Router IDs
- Finding OSPF Hello and Dead Timer Mismatches
- Shutting Down the OSPF Process
- Issues That Allow Adjacencies but Prevent IP Routes

- Mismatched MTU Settings
- Mismatched OSPF Network Types

IP Version 6

Chapter 22: Fundamentals of IP Version 6

- Introduction to IPv6
- The Historical Reasons for IPv6
- The IPv6 Protocols
- IPv6 Routing
- IPv6 Routing Protocols
- IPv6 Addressing Formats and Conventions
- Representing Full (Unabbreviated) IPv6 Addresses
- Abbreviating and Expanding IPv6 Addresses
- Abbreviating IPv6 Addresses
- Expanding Abbreviated IPv6 Addresses
- Representing the Prefix Length of an Address
- Calculating the IPv6 Prefix (Subnet ID)
- Finding the IPv6 Prefix
- Working with More Difficult IPv6 Prefix Lengths

Chapter 23: IPv6 Addressing and Subnetting

- Global Unicast Addressing Concepts
- Public and Private IPv6 Addresses
- The IPv6 Global Routing Prefix
- Address Ranges for Global Unicast Addresses
- IPv6 Subnetting Using Global Unicast Addresses
- Deciding Where IPv6 Subnets Are Needed
- The Mechanics of Subnetting IPv6 Global Unicast Addresses
- Listing the IPv6 Subnet Identifier
- List All IPv6 Subnets
- Assign Subnets to the Internetwork Topology

- Assigning Addresses to Hosts in a Subnet
- Unique Local Unicast Addresses
- Subnetting with Unique Local IPv6 Addresses
- The Need for Globally Unique Local Addresses

Chapter 24: Implementing IPv6 Addressing Routers

- Implementing Unicast IPv6 Addresses on Routers
- Static Unicast Address Configuration
- Configuring the Full 128-Bit Address
- Enabling IPv6 Routing
- Verifying the IPv6 Address Configuration
- Generating a Unique Interface ID Using Modified EUI-64
- Dynamic Unicast Address Configuration
- Special Addresses Used by Routers
- Link-Local Addresses
- Link-Local Address Concepts
- Creating Link-Local Addresses on Routers
- Routing IPv6 with Only Link-Local Addresses on an Interface
- IPv6 Multicast Addresses
- Reserved Multicast Addresses
- Multicast Address Scopes
- Solicited-Node Multicast Addresses
- Miscellaneous IPv6 Addresses
- Anycast Addresses
- IPv6 Addressing Configuration Summary

Chapter 25: Implementing IPv6 Routing

- Connected and Local IPv6 Routes
- Rules for Connected and Local Routes
- Example of Connected IPv6 Routes
- Examples of Local IPv6 Routes

- Static IPv6 Routes
- Static Routes Using the Outgoing Interface
- Static Routes Using Next-Hop IPv6 Address
- Example Static Route with a Global Unicast Next-Hop Address
- Example Static Route with a Link-Local Next-Hop Address
- Static Routes over Ethernet Links
- Static Default Routes
- Static IPv6 Host Routes
- Floating Static IPv6 Routes
- Troubleshooting Static IPv6 Routes
- Troubleshooting Incorrect Static Routes That Appear in the IPv6 Routing Table
- The Static Route Does Not Appear in the IPv6 Routing Table
- The Neighbor Discovery Protocol
- Discovering Neighbor Link Addresses with NDP NS and NA
- Discovering Routers with NDP RS and RA
- Using SLAAC with NDP RS and RA
- Discovering Duplicate Addresses Using NDP NS and NA
- NDP Summary

Wireless LANs

Chapter 26: Fundamentals of Wireless Networks

- Comparing Wired and Wireless Networks
- Wireless LAN Topologies
- Basic Service Set
- Distribution System
- Extended Service Set
- Independent Basic Service Set
- Other Wireless Topologies
- Repeater

- Workgroup Bridge
- Outdoor Bridge
- Mesh Network
- RF Overview
- Wireless Bands and Channels
- APs and Wireless Standards

Chapter 27: Analysing Cisco Wireless Architectures

- Autonomous AP Architecture
- Cloud-based AP Architecture
- Split-MAC Architectures
- Comparing Wireless LAN Controller Deployments
- Cisco AP Modes

Chapter 28: Securing Wireless Networks

- Anatomy of a Secure Connection
- Authentication
- Message Privacy
- Message Integrity
- Wireless Client Authentication Methods
- Open Authentication
- WEP
- 802.1x/EAP
- LEAP
- EAP-FAST
- PEAP
- EAP-TLS
- Wireless Privacy and Integrity Methods
- TKIP
- CCMP
- GCMP

- WPA, WPA2, and WPA3

Chapter 29: Building a Wireless LAN

- Connecting a Cisco AP
- Accessing a Cisco WLC
- Connecting a Cisco WLC
- Using WLC Ports
- Using WLC Interfaces
- Configuring a WLAN
- Step 1. Configure a RADIUS Server
- Step 2. Create a Dynamic Interface
- Step 3. Create a New WLAN
- Configuring WLAN Security
- Configuring WLAN QoS
- Configuring Advanced WLAN Settings
- Finalizing WLAN Configuration

IP Access Control Lists

Chapter 30: Introduction to TCP/IP Transport and Applications

- TCP/IP Layer 4 Protocols: TCP and UDP
- Transmission Control Protocol
- Multiplexing Using TCP Port Numbers
- Popular TCP/IP Applications
- Connection Establishment and Termination
- Error Recovery and Reliability
- Flow Control Using Windowing
- User Datagram Protocol
- TCP/IP Applications
- Uniform Resource Identifiers
- Finding the Web Server Using DNS
- Transferring Files with HTTP

- How the Receiving Host Identifies the Correct Receiving Application

Chapter 31: Basic IPv4 Access Control Lists

- IPv4 Access Control List Basics
- ACL Location and Direction
- Matching Packets
- Taking Action When a Match Occurs
- Types of IP ACLs
- Standard Numbered IPv4 ACLs
- List Logic with IP ACLs
- Matching Logic and Command Syntax
- Matching the Exact IP Address
- Matching a Subset of the Address with Wildcards
- Binary Wildcard Masks
- Finding the Right Wildcard Mask to Match a Subnet
- Matching Any/All Addresses
- Implementing Standard IP ACLs
- Standard Numbered ACL Example 1
- Standard Numbered ACL Example 2
- Troubleshooting and Verification Tip
- Practice Applying Standard IP ACLs
- Practice Building access-list Commands
- Reverse Engineering from ACL to Address Range

Chapter 32: Advanced IPv4 Access Control Lists

- Extended Numbered IP Access Control Lists
- Matching the Protocol, Source IP, and Destination IP
- Matching TCP and UDP Port Numbers
- Extended IP ACL Configuration
- Extended IP Access Lists: Example 1
- Extended IP Access Lists: Example 2

- Practice Building access-list Commands
- Named ACLs and ACL Editing
- Named IP Access Lists
- Editing ACLs Using Sequence Numbers
- Numbered ACL Configuration Versus Named ACL Configuration
- ACL Implementation Considerations
- Additional Reading on ACLs

Security Services

Chapter 33: Security Architectures

- Security Terminology
- Common Security Threats
- Attacks That Spoof Addresses
- Denial-of-Service Attacks
- Reflection and Amplification Attacks
- Man-in-the-Middle Attacks
- Address Spoofing Attack Summary
- Reconnaissance Attacks
- Buffer Overflow Attacks
- Malware
- Human Vulnerabilities
- Password Vulnerabilities
- Password Alternatives
- Controlling and Monitoring User Access
- Developing a Security Program to Educate Users

Chapter 34: Securing Network Devices

- Securing IOS Passwords
- Encrypting Older IOS Passwords with service password-encryption
- Encoding the Enable Passwords with Hashes
- Interactions Between Enable Password and Enable Secret

- Making the Enable Secret Truly Secret with a Hash
- Improved Hashes for Cisco's Enable Secret
- Encoding the Passwords for Local Usernames
- Controlling Password Attacks with ACLs
- Firewalls and Intrusion Prevention Systems
- Traditional Firewalls
- Security Zones
- Intrusion Prevention Systems (IPS)
- Cisco Next-Generation Firewalls
- Cisco Next-Generation IPS

Chapter 35: Implementing Switch Port Security

- Port Security Concepts and Configuration
- Configuring Port Security
- Verifying Port Security
- Port Security MAC Addresses
- Port Security Violation Modes
- Port Security Shutdown Mode
- Port Security Protect and Restrict Modes

Chapter 36: Implementing DHCP

- Dynamic Host Configuration Protocol
- DHCP Concepts
- Supporting DHCP for Remote Subnets with DHCP Relay
- Information Stored at the DHCP Server
- Configuring DHCP Features on Routers and Switches
- Configuring DHCP Relay
- Configuring a Switch as DHCP Client
- Configuring a Router as DHCP Client
- Identifying Host IPv4 Settings
- Host Settings for IPv4

- Host IP Settings on Windows
- Host IP Settings on macOS
- Host IP Settings on Linux

Chapter 37: DHCP Snooping and ARP Inspection

- DHCP Snooping Concepts
- A Sample Attack: A Spurious DHCP Server
- DHCP Snooping Logic
- Filtering DISCOVER Messages Based on MAC Address
- Filtering Messages that Release IP Addresses
- DHCP Snooping Configuration
- Configuring DHCP Snooping on a Layer 2 Switch
- Limiting DHCP Message Rates
- DHCP Snooping Configuration Summary
- Dynamic ARP Inspection
- DAI Concepts
- Review of Normal IP ARP
- Gratuitous ARP as an Attack Vector
- Dynamic ARP Inspection Logic
- Dynamic ARP Inspection Configuration
- Configuring ARP Inspection on a Layer 2 Switch
- Limiting DAI Message Rates
- Configuring Optional DAI Message Checks
- IP ARP Inspection Configuration Summary

IP Services

Chapter 38: Device Management Protocols

- System Message Logging (Syslog)
- Sending Messages in Real Time to Current Users
- Storing Log Messages for Later Review
- Log Message Format

- Log Message Severity Levels
- Configuring and Verifying System Logging
- The debug Command and Log Messages
- Network Time Protocol (NTP)
- Setting the Time and Timezone
- Basic NTP Configuration
- NTP Reference Clock and Stratum
- Redundant NTP Configuration
- NTP Using a Loopback Interface for Better Availability
- Analyzing Topology Using CDP and LLDP
- Examining Information Learned by CDP
- Configuring and Verifying CDP
- Examining Information Learned by LLDP
- Configuring and Verifying LLDP

Chapter 39: Network Address Translation

- Perspectives on IPv4 Address Scalability
- CIDR
- Private Addressing
- Network Address Translation Concepts
- Static NAT
- Dynamic NAT
- Overloading NAT with Port Address Translation
- NAT Configuration and Troubleshooting
- Static NAT Configuration
- Dynamic NAT Configuration
- Dynamic NAT Verification
- NAT Overload (PAT) Configuration
- NAT Troubleshooting

Chapter 40: Quality of Service (QoS)

- Introduction to QoS
- QoS: Managing Bandwidth, Delay, Jitter, and Loss
- Types of Traffic
- Data Applications
- Voice and Video Applications
- QoS, as mentioned in The Book
- QoS on Switches and Routers
- Classification and Marking
- Classification Basics
- Matching (Classification) Basics
- Classification of Routers with ACLs and NBAR
- Marking IP DSCP and Ethernet CoS
- Marking the IP Header
- Marking the Ethernet 802.1Q Header
- Other Marking Fields
- Defining Trust Boundaries
- DiffServ Suggested Marking Values
- Expedited Forwarding (EF)
- Assured Forwarding (AF)
- Class Selector (CS)
- Guidelines for DSCP Marking Values
- Queuing
- Round-Robin Scheduling (Prioritization)
- Low Latency Queuing
- A Prioritization Strategy for Data, Voice, and Video
- Shaping and Policing
- Policing
- Where to Use Policing
- Shaping

- Setting a Good Shaping Time Interval for Voice and Video
- Congestion Avoidance
- TCP Windowing Basics
- Congestion Avoidance Tools

Chapter 41: Miscellaneous IP Services

- First Hop Redundancy Protocol
- The Need for Redundancy in Networks
- The Need for a First Hop Redundancy Protocol
- The Three Solutions for First-Hop Redundancy
- HSRP Concepts
- HSRP Failover
- HSRP Load Balancing
- Simple Network Management Protocol
- SNMP Variable Reading and Writing: SNMP Get and Set
- SNMP Notifications: Traps and Informs
- The Management Information Base
- Securing SNMP FTP and TFTP
- Managing Cisco IOS Images with
- FTP/TFTP
- The IOS File System
- Upgrading IOS Images
- Copying a New iOS Image to a Local iOS File System Using TFTP
- Verifying IOS Code Integrity with MD5
- Copying Images with FTP
- The FTP and TFTP Protocols
- FTP Protocol Basics
- FTP Active and Passive Modes
- FTP over TLS (FTP Secure)
- TFTP Protocol Basics

Network Architecture

Chapter 42: LAN Architecture

- Analyzing Campus LAN Topologies
- Two-Tier Campus Design (Collapsed Core)
- The Two-Tier Campus Design
- Topology Terminology Seen Within a Two-Tier Design
- Three-Tier Campus Design (Core)
- Topology Design Terminology
- Small Office/Home Office
- Power over Ethernet (PoE)
- PoE Basics
- PoE Operation
- PoE and LAN Design

Chapter 43: WAN Architecture

- Metro Ethernet
- Metro Ethernet Physical Design and Topology
- Ethernet WAN Services and Topologies
- Ethernet Line Service (Point-to-Point)
- Ethernet LAN Service (Full Mesh)
- Ethernet Tree Service (Hub and Spoke)
- Layer 3 Design Using Metro Ethernet
- Layer 3 Design with E-Line Service
- Layer 3 Design with E-LAN Service
- Multiprotocol Label Switching (MPLS)
- MPLS VPN Physical Design and Topology
- MPLS and Quality of Service
- Layer 3 with MPLS VPN
- Internet VPNs
- Internet Access

- Digital Subscriber Line
- Cable Internet
- Wireless WAN (3G, 4G, LTE, 5G)
- Fiber (Ethernet) Internet Access
- Internet VPN Fundamentals
- Site-to-Site VPNs with IPsec
- Remote Access VPNs with TLS
- VPN Comparisons

Chapter 44: Cloud Architecture

- Server Virtualization
- Cisco Server Hardware
- Server Virtualization Basics
- Networking with Virtual Switches on a
- Virtualized Host
- The Physical Data Center Network
- Workflow with a Virtualized Data Center
- Cloud Computing Services
- Private Cloud (On-Premise)
- Public Cloud
- Cloud and the “Software as a Service” Model
- Infrastructure as a Service
- Software as a Service
- (Development) Platform as a Service
- WAN Traffic Paths to Reach Cloud Services
- Enterprise WAN Connections to Public Cloud
- Accessing Public Cloud Services Using the Internet
- Pros and Cons of Connecting to Public Cloud with Internet
- Private WAN and Internet VPN Access to Public Cloud
- Pros and Cons of Connecting to the Cloud with Private WANs

- Intercloud Exchanges
- Summarizing the Pros and Cons of Public Cloud WAN Options
- A Scenario: Branch Offices and the Public Cloud
- Migrating Traffic Flows When Migrating to Email SaaS
- Branch Offices with Internet and Private WAN

Network Automation

Chapter 45: Introduction to Controller-Based Networking

- SDN and Controller-Based Networks
- The Data, Control, and Management Planes
- The Data Plane
- The Control Plane
- The Management Plane
- Cisco Switch Data Plane Internals
- Controllers and Software-Defined Architecture
- Controllers and Centralized Control
- The Southbound Interface
- The Northbound Interface
- Software Defined Architecture Summary
- Examples of Network Programmability and SDN
- OpenDaylight and OpenFlow
- The OpenDaylight Controller
- The Cisco Open SDN Controller (OSC)
- Cisco Application Centric Infrastructure (ACI)
- ACI Physical Design: Spine and Leaf
- ACI Operating Model with Intent-Based Networking
- Cisco APIC Enterprise Domain
- APIC-EM Basics
- APIC-EM Replacement
- Summary of the SDN Examples

- Comparing Traditional Versus Controller-Based Networks
- How Automation Impacts Network Management
- Comparing Traditional Networks with Controller-Based Networks

Chapter 46: Cisco Software-Defined Access (SDA)

- SDA Fabric, Underlay, and Overlay
- The SDA Underlay
- Using Existing Gear for the SDA Underlay
- Using New Gear for the SDA Underlay
- The SDA Overlay
- VXLAN Tunnels in the Overlay (Data Plane)
- LISP for Overlay Discovery and Location (Control Plane)
- DNA Center and SDA Operation
- Cisco DNA Center
- Cisco DNA Center and Scalable Groups
- Issues with Traditional IP-Based Security
- SDA Security Based on User Groups
- DNA Center as a Network Management Platform
- DNA Center Similarities to Traditional Management
- DNA Center Differences with Traditional Management

Chapter 47: Understanding REST and JSON

- REST-Based APIs
- REST-Based (RESTful) APIs
- Client/Server Architecture
- Stateless Operation
- Cacheable (or not)
- Background: Data and Variables
- Simple Variables
- List and Dictionary Variables
- REST APIs and HTTP

- Software CRUD Actions and HTTP Verbs
- Using URIs with HTTP to Specify the Resource
- Example of REST API Call to DNA Center
- Data Serialization and JSON
- The Need for a Data Model with APIs
- Data Serialization Languages
- JSON
- XML
- YAML
- Summary of Data Serialization
- Interpreting JSON
- Interpreting JSON Key: Value Pairs
- Interpreting JSON Objects and Arrays
- Minified and Beautified JSON

Chapter 48: Understanding Ansible, Puppet, and Chef

- Device Configuration Challenges and Solutions
- Configuration Drift
- Centralized Configuration Files and Version Control
- Configuration Monitoring and Enforcement
- Configuration Provisioning
- Configuration Templates and Variables
- Files That Control Configuration Automation
- Ansible, Puppet, and Chef Basics
- Ansible
- Puppet
- Chef
- Summary of Configuration Management Tool

CCNP ENCOR- Module 1: Enterprise Network Architecture

Topics Covered

- Enterprise Campus Architecture
- Three-Tier Architecture
- Spine-Leaf Architecture
- High Availability Design
- Redundancy Models
- FHRP Overview
- Stateful Switch Over (SSO)
- Enterprise Network Design Principles
- Capacity Planning

Practical Labs

- Design a hierarchical enterprise network
- Configure high availability architecture

Learning Outcomes

- Design scalable enterprise networks
 - Implement redundancy and resiliency solutions
-

Module 2: Software Defined Architecture

Topics Covered

- Software Defined Networking (SDN)
- Cisco SD-Access Architecture
- Cisco SD-WAN Architecture
- Control Plane vs Data Plane
- DNA Center Fundamentals
- Fabric Networks
- Underlay and Overlay Networks

Practical Labs

- Analyze SD-WAN topology
- SD-Access fabric design exercises

Learning Outcomes

- Understand Cisco's modern enterprise architectures
 - Compare traditional and software-defined networks
-

Module 3: Network Virtualization

Topics Covered

- Virtual Machines
- Hypervisors (Type 1 & Type 2)
- Device Virtualization
- VRF Lite
- Overlay Technologies
- GRE Tunnels
- IPSec Tunnels
- VXLAN Fundamentals
- LISP Overview

Practical Labs

- Configure VRF Lite
- Configure GRE Tunnels
- Implement VXLAN concepts

Learning Outcomes

- Deploy virtualization technologies
 - Separate traffic using virtual routing instances
-

Module 4: Advanced Layer 2 Technologies

Topics Covered

- VLAN Design
- 802.1Q Trunking

- EtherChannel
- LACP and PAgP
- Rapid STP (RSTP)
- Multiple STP (MST)
- STP Optimization
- Root Guard
- BPDU Guard
- Loop Prevention

Practical Labs

- Configure EtherChannel
- Troubleshoot STP
- Configure MST

Learning Outcomes

- Implement resilient Layer 2 networks
 - Troubleshoot switching issues
-

Module 5: Advanced Routing Concepts

Topics Covered

- Routing Fundamentals Review
- Administrative Distance
- Route Selection
- Route Redistribution Concepts
- Policy-Based Routing (PBR)
- Route Maps
- Prefix Lists
- Bidirectional Forwarding Detection (BFD)

Practical Labs

- Configure PBR
- Implement route filtering

Learning Outcomes

- Control traffic paths
 - Implement advanced routing policies
-

Module 6: OSPF Enterprise Routing

Topics Covered

- OSPFv2 and OSPFv3
- Multi-Area OSPF
- LSA Types
- DR/BDR Election
- OSPF Authentication
- Route Summarization
- Stub Areas
- NSSA
- OSPF Troubleshooting

Practical Labs

- Configure Multi-Area OSPF
- Configure Stub Areas
- Troubleshoot OSPF Adjacencies

Learning Outcomes

- Deploy enterprise OSPF environments
 - Optimize OSPF routing design
-

Module 7: EIGRP Enterprise Routing

Topics Covered

- EIGRP Operations
- DUAL Algorithm
- Neighbor Relationships
- Metric Calculation

- Route Summarization
- Stub Routing
- Authentication
- Wide Metrics

Practical Labs

- Configure EIGRP
- Verify DUAL Operations

Learning Outcomes

- Implement and troubleshoot EIGRP networks
-

Module 8: BGP Fundamentals for Enterprise Networks

Topics Covered

- eBGP Configuration
- iBGP Configuration
- BGP Attributes
- Best Path Selection
- Route Advertisement
- Route Filtering
- Communities
- AS Path Manipulation
- Multi-Homing Design

Practical Labs

- Configure eBGP
- Manipulate BGP Path Selection

Learning Outcomes

- Deploy enterprise BGP solutions
 - Control routing decisions
-

Module 9: Multicast Technologies

Topics Covered

- Multicast Fundamentals
- IGMP v2/v3
- IGMP Snooping
- PIM Dense Mode
- PIM Sparse Mode
- Source Specific Multicast (SSM)
- Bidirectional PIM
- MSDP
- Reverse Path Forwarding (RPF)

Practical Labs

- Configure PIM Sparse Mode
- Verify Multicast Trees

Learning Outcomes

- Deploy scalable multicast solutions
-

Module 10: Enterprise IP Services

Topics Covered

- NAT
- PAT
- HSRP
- VRRP
- GLBP
- NTP
- Precision Time Protocol (PTP)
- DHCP Services
- DNS Fundamentals

Practical Labs

- Configure HSRP

- Configure NAT/PAT
- Configure NTP

Learning Outcomes

- Implement enterprise network services
-

Module 11: Quality of Service (QoS)

Topics Covered

- QoS Models
- Traffic Classification
- Marking and Trust Boundaries
- DSCP and CoS
- Congestion Management
- Queuing Mechanisms
- Policing
- Shaping
- Low Latency Queuing (LLQ)

Practical Labs

- Configure QoS Policies
- Implement Traffic Shaping

Learning Outcomes

- Prioritize business-critical traffic
-

Module 12: Wireless Enterprise Fundamentals

Topics Covered

- Wireless Architectures
- WLAN Design
- Wireless Controllers
- CAPWAP
- Wireless Roaming

- RF Fundamentals
- Wireless Troubleshooting
- Wireless Deployment Models

Practical Labs

- Configure Wireless LAN Controllers
- Analyze WLAN Design

Learning Outcomes

- Understand enterprise wireless infrastructure
-

Module 13: Network Assurance and Monitoring

Topics Covered

- SNMP
- Syslog
- Flexible NetFlow
- SPAN
- RSPAN
- ERSPAN
- IP SLA
- Performance Monitoring
- Troubleshooting Methodologies

Practical Labs

- Configure NetFlow
- Configure SPAN Sessions
- Configure IP SLA

Learning Outcomes

- Monitor network performance and availability
-

Module 14: Infrastructure Security

Topics Covered

- Device Hardening
- Secure Management Access
- SSH
- AAA
- TACACS+
- RADIUS
- 802.1X
- Network Access Control
- Control Plane Policing (CoPP)
- Management Plane Protection (MPP)

Practical Labs

- Configure AAA
- Configure 802.1X
- Implement CoPP

Learning Outcomes

- Secure enterprise infrastructure
-

Module 15: Enterprise Security Technologies

Topics Covered

- Access Control Lists
- IPSec VPN
- GRE over IPSec
- MACsec
- TrustSec
- Web Authentication
- MAC Authentication Bypass (MAB)
- Next-Generation Firewall Concepts
- Endpoint Security

Practical Labs

- Configure IPsec VPN
- Configure ACL Policies

Learning Outcomes

- Implement enterprise-grade security controls
-

Module 16: Automation and Programmability

Topics Covered

- Network Automation Fundamentals
- Python Basics for Networking
- JSON
- YAML
- REST APIs
- RESTCONF
- NETCONF
- YANG Data Models
- Cisco DNA APIs
- Cisco vManage APIs

Practical Labs

- Parse JSON with Python
- Use REST APIs
- Automate device configurations

Learning Outcomes

- Automate network operations and management
-

Module 17: AI and Modern Network Operations

Topics Covered

- AI-Assisted Network Management
- Cisco AI Features
- Intent-Based Networking

- Predictive Analytics
- Automated Troubleshooting
- Telemetry and Observability

Practical Labs

- Analyze AI-driven network workflows
- Telemetry data interpretation

Learning Outcomes

- Understand AI integration in enterprise networking
-

Module 18: Enterprise Troubleshooting Workshop

Topics Covered

- Layer 2 Troubleshooting
- Routing Troubleshooting
- OSPF Issues
- BGP Issues
- Wireless Troubleshooting
- Security Troubleshooting
- Performance Analysis

Practical Labs

- Real-world troubleshooting scenarios
- Enterprise fault isolation exercises

Learning Outcomes

- Apply systematic troubleshooting methodologies
-

Module 19: CCNP ENCOR Enterprise Lab

Hands-On Activities

- Configure Enterprise Campus Network
- Implement OSPF and BGP
- Configure Multicast

- Configure QoS
- Implement AAA
- Configure VPNs
- Implement Automation Scripts
- Troubleshoot Complex Scenarios

Deliverables

- Configuration Files
 - Network Diagrams
 - Verification Reports
-

Module 20: Enterprise Network Design Capstone

End-to-End Enterprise Deployment

Project Tasks

- Design Campus Architecture
- Create Routing Architecture
- Implement Security Controls
- Configure High Availability
- Deploy Monitoring Systems
- Implement Automation Framework

Deliverables

- Enterprise Network Design Document
- Routing Design
- Security Architecture
- Automation Scripts
- Monitoring Dashboard

Module 1: Advanced Enterprise Routing Architecture

Topics Covered

- Enterprise Routing Design Principles
- Routing Table Processing
- Forwarding Information Base (FIB)
- Cisco Express Forwarding (CEF)
- Control Plane vs Data Plane
- Routing Scalability
- High Availability Design

Practical Labs

- Analyze routing table behavior
- Verify CEF operations

Learning Outcomes

- Understand enterprise routing architecture.
 - Analyze packet forwarding decisions.
-

Module 2: Advanced OSPF

Topics Covered

- OSPFv2 Deep Dive
- OSPFv3 Deep Dive
- LSA Types and Functions
- Multi-Area OSPF
- Stub Areas
- Totally Stubby Areas
- NSSA
- Virtual Links
- OSPF Summarization
- OSPF Authentication
- Fast Convergence Techniques

Practical Labs

- Configure Multi-Area OSPF
- Implement NSSA and Stub Areas
- Troubleshoot OSPF Adjacencies

Learning Outcomes

- Deploy scalable OSPF networks.
 - Troubleshoot OSPF routing issues.
-

Module 3: Advanced EIGRP

Topics Covered

- DUAL Algorithm
- Feasible Successor Selection
- Route Summarization
- Stub Routing
- Authentication
- Wide Metrics
- Named EIGRP Configuration
- Load Balancing
- Query Boundaries

Practical Labs

- Configure Named EIGRP
- Implement EIGRP Authentication
- Analyze DUAL Operations

Learning Outcomes

- Optimize EIGRP deployments.
 - Troubleshoot EIGRP convergence issues.
-

Module 4: Advanced BGP

Topics Covered

- eBGP and iBGP
- BGP Neighbor Relationships
- BGP Path Selection Process
- BGP Attributes
- Local Preference
- MED
- Weight
- Communities
- Route Reflectors
- Confederations
- Route Aggregation
- Route Filtering
- Prefix Lists
- Route Maps

Practical Labs

- Configure Enterprise BGP
- Manipulate Best Path Selection
- Configure Route Reflectors

Learning Outcomes

- Deploy enterprise BGP architectures.
- Implement routing policies.

Module 5: Route Redistribution

Topics Covered

- Redistribution Concepts
- OSPF to EIGRP Redistribution
- EIGRP to OSPF Redistribution
- BGP Redistribution
- Seed Metrics

- Route Tagging
- Redistribution Loops
- Filtering Redistribution

Practical Labs

- Configure Mutual Redistribution
- Prevent Routing Loops

Learning Outcomes

- Implement route sharing across protocols.
 - Troubleshoot redistribution problems.
-

Module 6: Route Manipulation and Policy Control

Topics Covered

- Route Maps
- Prefix Lists
- Distribute Lists
- Administrative Distance Manipulation
- Policy-Based Routing (PBR)
- Object Tracking
- Conditional Routing
- Traffic Engineering

Practical Labs

- Configure PBR
- Implement Route Filtering

Learning Outcomes

- Control traffic flow and routing decisions.
-

Module 7: MPLS Fundamentals

Topics Covered

- MPLS Architecture

- Label Switching
- LER and LSR
- MPLS Labels
- Label Distribution Protocol (LDP)
- MPLS VPN Concepts
- Service Provider Fundamentals

Practical Labs

- MPLS Traffic Flow Analysis
- Label Distribution Verification

Learning Outcomes

- Understand MPLS operations.
 - Analyze MPLS-based networks.
-

Module 8: VPN Technologies

Topics Covered

- Site-to-Site VPN
- GRE Tunnels
- IPSec VPN
- DMVPN
- FlexVPN
- Remote Access VPN
- VPN Design Considerations
- Encryption Fundamentals

Practical Labs

- Configure GRE Tunnel
- Configure IPSec VPN
- Implement DMVPN

Learning Outcomes

- Secure enterprise communications.

- Deploy VPN technologies.
-

Module 9: Infrastructure Security

Topics Covered

- AAA Architecture
- TACACS+
- RADIUS
- Device Hardening
- Control Plane Protection (CoPP)
- Management Plane Protection
- Secure Device Access
- Role-Based Access Control (RBAC)

Practical Labs

- Configure AAA
- Implement TACACS+
- Secure Router Access

Learning Outcomes

- Strengthen network infrastructure security.
-

Module 10: First Hop Redundancy Protocols (FHRP)

Topics Covered

- HSRP
- VRRP
- GLBP
- Object Tracking
- FHRP Optimization
- Load Balancing Strategies

Practical Labs

- Configure HSRP

- Configure GLBP
- Verify Failover Operations

Learning Outcomes

- Implement gateway redundancy solutions.
-

Module 11: Enterprise Infrastructure Services

Topics Covered

- DHCP
- DHCP Relay
- DNS
- NTP
- SNMPv3
- Syslog
- SLA Operations
- Infrastructure Monitoring

Practical Labs

- Configure DHCP Relay
- Configure NTP
- Configure SNMPv3

Learning Outcomes

- Deploy critical enterprise services.
-

Module 12: Quality of Service (QoS)

Topics Covered

- QoS Models
- Classification and Marking
- DSCP
- CoS
- Policing

- Traffic Shaping
- Congestion Avoidance
- Low Latency Queuing

Practical Labs

- Configure QoS Policies
- Verify Traffic Prioritization

Learning Outcomes

- Optimize network performance.
-

Module 13: Multicast Routing

Topics Covered

- Multicast Concepts
- IGMP
- IGMP Snooping
- PIM Dense Mode
- PIM Sparse Mode
- RP Configuration
- Source Specific Multicast (SSM)
- Reverse Path Forwarding (RPF)

Practical Labs

- Configure PIM-SM
- Verify Multicast Trees

Learning Outcomes

- Deploy multicast solutions.
-

Module 14: Enterprise Network Monitoring

Topics Covered

- NetFlow
- Flexible NetFlow

- SPAN
- RSPAN
- ERSPAN
- IP SLA
- SNMP Monitoring
- Telemetry Concepts

Practical Labs

- Configure NetFlow
- Configure SPAN
- Implement IP SLA

Learning Outcomes

- Monitor and troubleshoot network performance.
-

Module 15: Troubleshooting Enterprise Networks

Topics Covered

- Structured Troubleshooting Methodology
- OSPF Troubleshooting
- EIGRP Troubleshooting
- BGP Troubleshooting
- VPN Troubleshooting
- QoS Troubleshooting
- Security Troubleshooting

Practical Labs

- Real-world fault isolation exercises
- Routing protocol troubleshooting

Learning Outcomes

- Resolve enterprise network issues efficiently.
-

Module 16: Advanced Troubleshooting Workshop

Topics Covered

- Multi-Protocol Routing Failures
- Redistribution Issues
- VPN Connectivity Problems
- High Availability Failures
- Route Filtering Problems
- Convergence Analysis

Practical Labs

- Complex enterprise troubleshooting scenarios

Learning Outcomes

- Diagnose advanced networking problems.
-

Module 17: Enterprise Routing Lab Workshop

Hands-On Activities

- Multi-Area OSPF Deployment
- EIGRP Optimization
- Enterprise BGP Configuration
- Route Redistribution
- VPN Deployment
- FHRP Configuration
- QoS Policy Creation

Deliverables

- Configuration Files
 - Routing Verification Reports
 - Troubleshooting Documentation
-

Module 18: Enterprise Routing Capstone Project

End-to-End Enterprise WAN Deployment

Project Tasks

- Design Enterprise Routing Architecture
- Implement OSPF, EIGRP, and BGP
- Configure Route Redistribution
- Deploy VPN Technologies
- Implement Security Controls
- Configure Monitoring Solutions
- Validate High Availability

Deliverables

- Network Design Document
- Routing Architecture Diagram
- Security Implementation Plan
- VPN Design Guide
- Troubleshooting Runbook

SailPoint IIQ

Module 1: Introduction to Identity and Access Management (IAM)

Topics Covered

- Fundamentals of Identity and Access Management
- Authentication vs Authorization
- Identity Governance and Administration (IGA)
- Identity Lifecycle Management
- IAM Components and Architecture
- Access Control Models
 - RBAC
 - ABAC

- PBAC
- Regulatory Compliance Requirements

Learning Outcomes

- Understand IAM fundamentals.
 - Explain the role of Identity Governance in enterprises.
-

Module 2: Introduction to SailPoint IdentityIQ

Topics Covered

- IdentityIQ Overview
- IdentityIQ Architecture
- Core Components
- Identity Cube Concept
- Governance Framework
- SailPoint Product Ecosystem
- IdentityIQ vs Identity Security Cloud

Practical Labs

- Navigate IdentityIQ Interface
- Explore Administrative Console

Learning Outcomes

- Understand IdentityIQ architecture and functionality.
 - Navigate the IdentityIQ environment.
-

Module 3: IdentityIQ Architecture and Components

Topics Covered

- Application Server Architecture
- Database Components
- Identity Warehouse
- Identity Cubes
- Aggregation Engine

- Provisioning Engine
- Workflow Engine
- Task Scheduler
- Audit Framework

Practical Labs

- Analyze IdentityIQ architecture
- Review identity warehouse structure

Learning Outcomes

- Understand how IdentityIQ components interact.
-

Module 4: Identity Lifecycle Management

Topics Covered

- Joiner Process
- Mover Process
- Leaver Process
- Identity Creation
- Identity Updates
- Identity Termination
- Automated Lifecycle Events
- Lifecycle Event Triggers

Practical Labs

- Configure lifecycle events
- Test identity onboarding workflows

Learning Outcomes

- Automate identity lifecycle processes.
-

Module 5: Application Onboarding and Connectors

Topics Covered

- Application Integration Concepts

- Direct Connectors
- Delimited File Connectors
- JDBC Connectors
- Active Directory Integration
- LDAP Integration
- SAP Integration
- Cloud Application Integration

Practical Labs

- Configure Active Directory Connector
- Perform Application Aggregation

Learning Outcomes

- Integrate enterprise applications with IdentityIQ.
-

Module 6: Identity Aggregation and Correlation

Topics Covered

- Identity Aggregation
- Account Aggregation
- Correlation Rules
- Identity Mapping
- Identity Cube Creation
- Data Normalization
- Aggregation Troubleshooting

Practical Labs

- Configure correlation rules
- Perform account aggregation

Learning Outcomes

- Build and maintain accurate identity repositories.
-

Module 7: Access Request Management

Topics Covered

- Access Request Framework
- Shopping Cart Model
- Request Approval Workflows
- Multi-Level Approvals
- Access Fulfillment
- Self-Service Access Requests
- Request Auditing

Practical Labs

- Create access request workflows
- Configure approval chains

Learning Outcomes

- Manage access requests and approvals.
-

Module 8: Provisioning and Deprovisioning

Topics Covered

- Provisioning Concepts
- Provisioning Policies
- Automated Provisioning
- Birthright Access
- Deprovisioning Workflows
- Provisioning Plans
- Provisioning Troubleshooting

Practical Labs

- Configure provisioning policies
- Test automated provisioning

Learning Outcomes

- Automate access assignment and removal.
-

Module 9: Role Management

Topics Covered

- Business Roles
- IT Roles
- Role Hierarchies
- Role Mining
- Role Modeling
- Role Assignment
- Role Governance

Practical Labs

- Create business roles
- Build role hierarchies

Learning Outcomes

- Design scalable role-based access models.
-

Module 10: Access Certifications and Reviews

Topics Covered

- Certification Concepts
- Manager Certifications
- Application Owner Certifications
- Role Certifications
- Identity Certifications
- Campaign Management
- Certification Remediation

Practical Labs

- Create certification campaigns
- Execute access reviews

Learning Outcomes

- Conduct governance-based access reviews.

Module 11: Policy Management and Compliance

Topics Covered

- Segregation of Duties (SoD)
- Policy Violations
- Risk Scoring
- Compliance Monitoring
- Policy Enforcement
- Regulatory Mapping
- Audit Readiness

Practical Labs

- Configure SoD Policies
- Analyze policy violations

Learning Outcomes

- Implement governance and compliance controls.
-

Module 12: Workflow Management

Topics Covered

- Workflow Fundamentals
- Workflow Components
- Approval Workflows
- Provisioning Workflows
- Escalation Workflows
- Workflow Variables
- Workflow Debugging

Practical Labs

- Create custom workflows
- Troubleshoot workflow execution

Learning Outcomes

- Design and manage IdentityIQ workflows.
-

Module 13: Rules and Customization

Topics Covered

- Rule Framework
- BeanShell Scripting
- Rule Types
 - Correlation Rules
 - Provisioning Rules
 - Validation Rules
 - Policy Rules
- Custom Logic Development
- Debugging Rules

Practical Labs

- Create custom rules
- Debug rule execution

Learning Outcomes

- Extend IdentityIQ functionality through customization.
-

Module 14: Reporting and Auditing

Topics Covered

- Audit Logs
- Compliance Reports
- Certification Reports
- Provisioning Reports
- Custom Reports
- Report Scheduling
- Governance Dashboards

Practical Labs

- Generate audit reports
- Build custom reports

Learning Outcomes

- Monitor governance activities effectively.
-

Module 15: IdentityIQ Administration

Topics Covered

- System Configuration
- Task Management
- Task Scheduling
- User Administration
- Backup and Recovery
- Performance Tuning
- Troubleshooting Techniques

Practical Labs

- Configure scheduled tasks
- Monitor system performance

Learning Outcomes

- Administer and maintain IdentityIQ environments.
-

Module 16: IdentityIQ Security

Topics Covered

- Security Architecture
- Authentication Integration
- Authorization Controls
- Encryption Mechanisms
- Secure Configuration Practices
- Access Control Policies
- Security Monitoring

Practical Labs

- Configure authentication methods
- Secure administrative access

Learning Outcomes

- Secure IdentityIQ deployments.
-

Module 17: IdentityIQ Advanced Development

Topics Covered

- Object Model
- XML Configuration
- Form Development
- Custom UI Development
- Plugin Development
- API Integration
- Advanced Troubleshooting

Practical Labs

- Develop custom forms
- Create custom plugins

Learning Outcomes

- Develop advanced IdentityIQ customizations.
-

Module 18: IdentityIQ Integration and APIs

Topics Covered

- REST APIs
- Web Services Integration
- SCIM Integration
- Third-Party Integrations
- HR System Integration
- ServiceNow Integration

- Enterprise Application Integration

Practical Labs

- Integrate external systems
- Configure API-based provisioning

Learning Outcomes

- Extend IdentityIQ through integrations.
-

Module 19: SailPoint IdentityIQ Implementation Workshop

Hands-On Activities

- Configure Identity Sources
- Integrate Active Directory
- Create Roles
- Configure Provisioning
- Launch Certification Campaigns
- Build Workflows
- Configure Policies

Deliverables

- Identity Governance Configuration
- Access Request Workflow
- Certification Campaign
- Governance Reports